17

Direct Gold Restorations

GREGORY E. SMITH

Direct gold is a gold restorative material that is manufactured for compaction directly into prepared cavities. Two types of direct gold are manufactured for dental use: gold foil and powdered gold. These gold materials differ in their metallurgic structure.

Pure gold has been in use in dentistry in the United States for more than 100 years.[1-6] Various techniques have been advanced for its use in the restoration of teeth. It is generally agreed that this noble metal is a superior restorative material for treatment of many small lesions and defects in teeth, given sound pulpal and periodontal health. Success is achieved with direct gold restorations if meticulous care is given to an exacting technique in tooth preparation design and material manipulation. Direct gold restorations can last for a lifetime if attention is paid to details of restorative technique and to proper home care. The longevity of direct gold restorations is a result of the superb biocompatibility of gold with the oral environment and its excellent marginal integrity.

This chapter discusses the various forms of direct gold presently available and explains the principles required for their manipulation. The principles of tooth preparation are reviewed as they are applied to direct gold restorations. Class I, V, and III preparations and their restoration are considered in detail.

Materials and Manufacture

Several physical types of direct-filling gold have been produced.[7] All are "compactable" in that they are inserted into tooth preparations under force and compacted or condensed into preparation line and point angles and against preparation walls.

The gold foil referred to in the restorative sections of this chapter is in pellet form (Figs. 17.1 and 17.2). Each piece is placed on clean fingertips, and the corners are tucked into the center (see Fig. 17.1B and C), and then the foil is lightly rolled into pellet form (see Fig. 17.1D). In addition, cylinders of gold foil may be rolled from the segments of a sheet (see Fig. 17.1A). After pellets of gold are rolled, they may be conveniently stored in a gold foil box (see Fig. 17.2), which is divided into labeled sections for various sizes of pellets. Cylinders of foil and selected sizes of other types of gold also may be stored in the box. Preferential contamination is suggested by placing a damp cotton pellet dipped into 18% ammonia into each section of the box. This serves to prevent deleterious oxides from forming on the gold until it is used.

Powdered gold is made by a combination of chemical precipitation and atomization, with an average particle size of 15 mm (Fig. 17.3A).[8] The atomized particles are mixed together in wax, cut into pieces, and wrapped in No. 4 or No. 3 foil (see Fig. 17.3B). Several sizes of these pellets are available. This product is marketed as Williams E-Z Gold (Ivoclar-Williams, Amherst, NY).

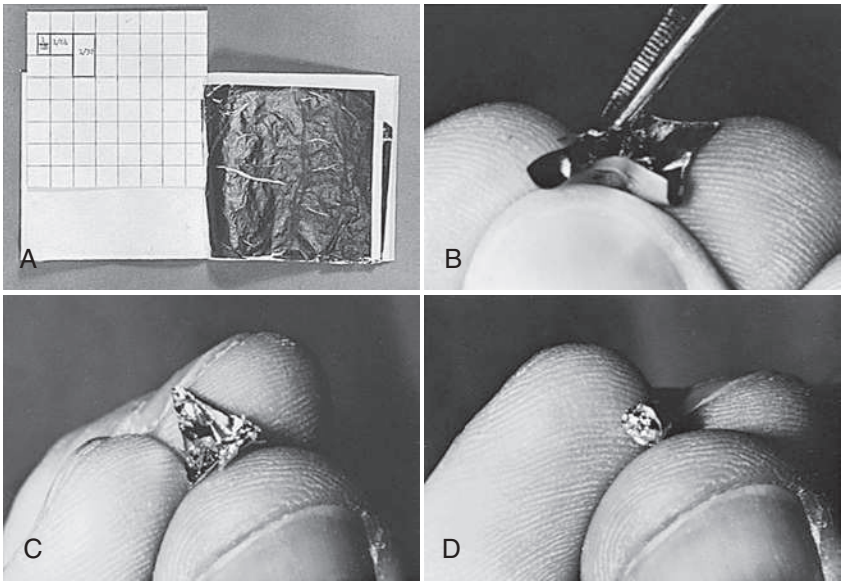
Cohesion and Degassing

Direct gold is inserted into tooth preparations under force. The purpose of the force is to weld the gold into restorations containing minimal porosity or internal void spaces.[9-11] Welding occurs because pure gold with an absolutely clean surface coheres as a result of metallic bonding. As the gold is forced and compressed into a tooth preparation, succeeding increments cohere to those previously placed. For successful welding to occur during restoration, the gold must be in a cohesive state before compaction, and a suitable, biologically compatible compacting force must be delivered.

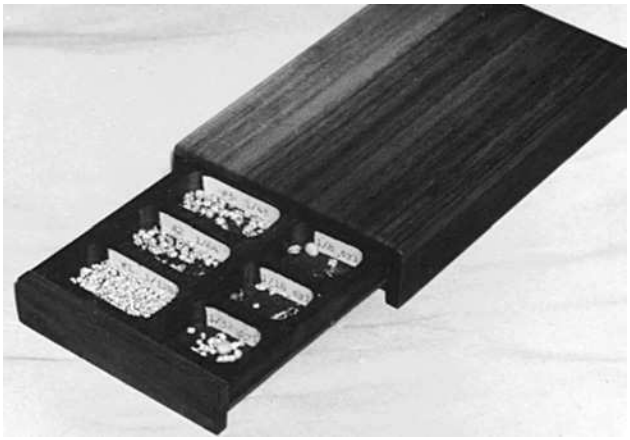
Direct gold may be either cohesive or noncohesive. It is noncohesive in the presence of surface impurities or wax, which prevents one increment of gold from cohering to another. The manufacturer supplies books of gold foil or prerolled cylinders in a cohesive or noncohesive state. E-Z Gold pellets are supplied with a wax coating that must be burned off before compaction.

Because gold attracts gases that render it noncohesive, such gases must be removed from the surface of the gold before dental compaction. This process usually is referred to as *degassing* or *annealing* and is accomplished by application of heat. The term *degassing* is preferable because the desired result is to remove residual surface contamination (although further annealing, resulting in additional internal stress relief or recrystallization, also may occur in this process). All direct-filling gold products are degassed immediately before use except when noncohesive foil is specifically desired. Underheating during degassing should be avoided because it fails to render the gold surface pure. Overheating also should be avoided because it may cause the gold to become brittle or melt and render it unusable. Degassing is accomplished by heating the gold foil on a mica tray over a flame or on an electric annealer or by heating each piece of gold over a pure ethanol flame (Fig. 17.4).

The advantage of the technique involving use of the pure ethanol flame is that each piece of gold is selected and heated just before insertion, and waste of gold is avoided. A careful technique is needed to degas an increment of gold in the flame correctly. The gold is passed into the blue inner core of the flame on the tip of a foil-passing instrument and held just until the gold becomes dull red, and then the instrument is withdrawn from the flame. After a few seconds are allowed for cooling, the gold is placed in the preparation. Although any of the three degassing procedures



• **Fig. 17.1** A, 10 × 10 cm (4 × 4 inch) book of foil marked for cutting and rolling into pellets of various sizes. B and C, Corners of foil piece are tucked into center. D, Foil is rolled into a completed pellet. (A, Courtesy Terkla and Cantwell.)



• **Fig. 17.2** Gold foil box. Compartments are labeled to show pellet size.

is satisfactory for gold foil, this is not the case for E-Z Gold. The E-Z Gold pellet must be heated ½ to 2.5 cm (1 inch) above the ethanol flame until a bright flame occurs (caused by ignition of the wax) and the pellet becomes dull red for 2 to 3 seconds, then it is withdrawn from above the flame.

Principles of Compaction

Direct-filling gold must be compacted during insertion into tooth preparations.[12] With the exception of E-Z Gold, the compaction takes the form of malleting forces that are delivered either by a hand mallet used by the assistant or by an Electro-Mallet (McShirley Products, Glendale, CA) or a pneumatic mallet used by the dentist. E-Z Gold, because of its powdered form, may be compacted by heavy hand pressure delivered in a rocking motion with specially designed hand condensers.[13,14] Successful malleting of the gold foil may be achieved with any of the currently available equipment. Some operators prefer the Electro-Mallet or the pneumatic mallet because a dental assistant is not required for the procedure.

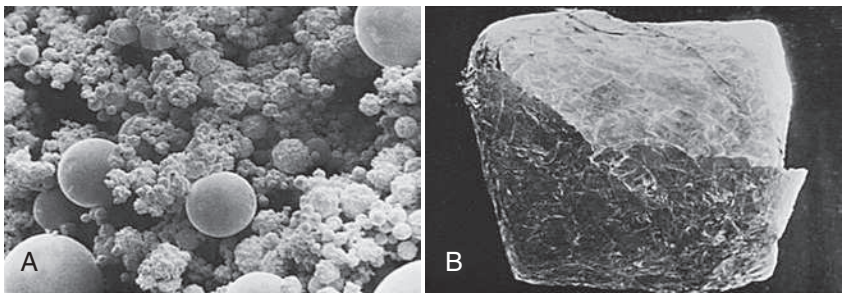
A technique preferred by many clinicians uses a hand mallet to deliver light blows to a condenser held by the dentist (Fig. 17.5A). This technique allows great control of malleting forces when variations are called for, and it allows for rapid change in condenser nibs, or tips, when a multitude of condensers is required. In any case, a suitable condenser must be stepped over the gold systematically to achieve a dense, well-compacted restoration (see Fig. 17.8 later in the chapter).

Condensers are designed to deliver forces of compaction to direct gold. Condensers used in the handpieces of the Electro-Mallet or pneumatic mallet consist of a nib, or working tip, and a short shank (approximately 2.5 cm in length) that fits into the malleting handpiece. Condensers used with the hand mallet are longer (approximately 15 cm) and have a blunt-ended handle that receives light blows from the hand mallet.

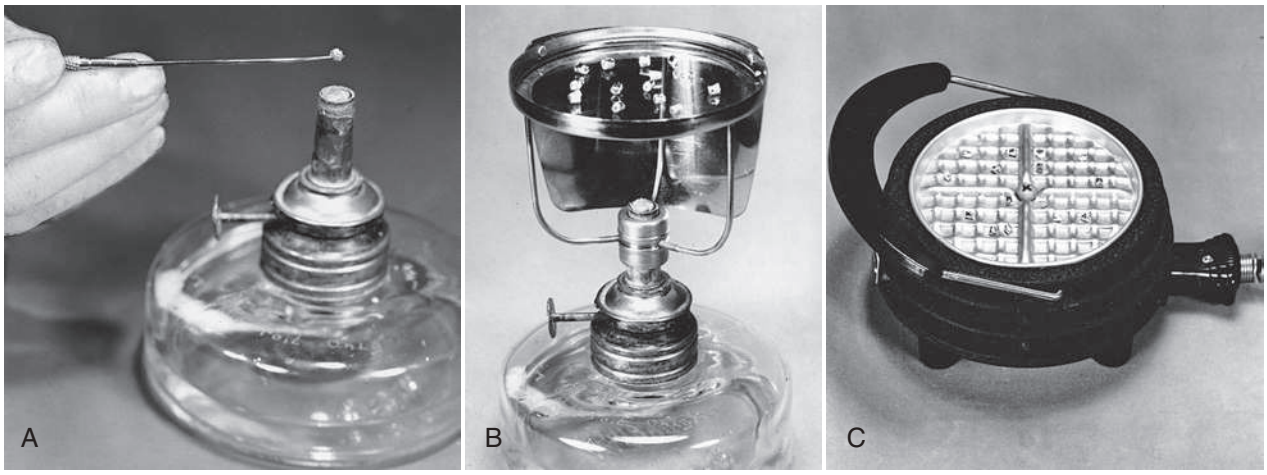
Condenser nibs are available in several shapes and sizes (see Fig. 17.5B). All have pyramidal serrations on the nib faces to prevent slipping on the gold. Condensers described in this chapter are (1) the round condensers, 0.4 to 0.55 mm in diameter; (2) the Varney foot condenser, which has a rectangular face that is approximately 1 to 1.3 mm; and (3) the parallelogram condensers, which are used only for hand pressure compaction and have nib faces that measure approximately 0.5 to 1 mm.

Condenser shanks may be straight, monangled, or offset, and their nib faces may be cut perpendicular to the long axis of the handle or perpendicular to the end portion of the shank (Fig. 17.6). The smaller the nib face size (i.e., area), the greater the pounds per square inch delivered (given a constant malleting force). If the nib diameter is reduced by half, the effective compaction force in pounds per square inch is four times greater (because the area of a circle is proportional to the square of the diameter). For most gold, the 0.4- to 0.55-mm diameter nibs are suitable. Smaller condensers tend to punch holes in the gold, whereas larger ones are less effective in forcing the gold into angles in the tooth preparation.

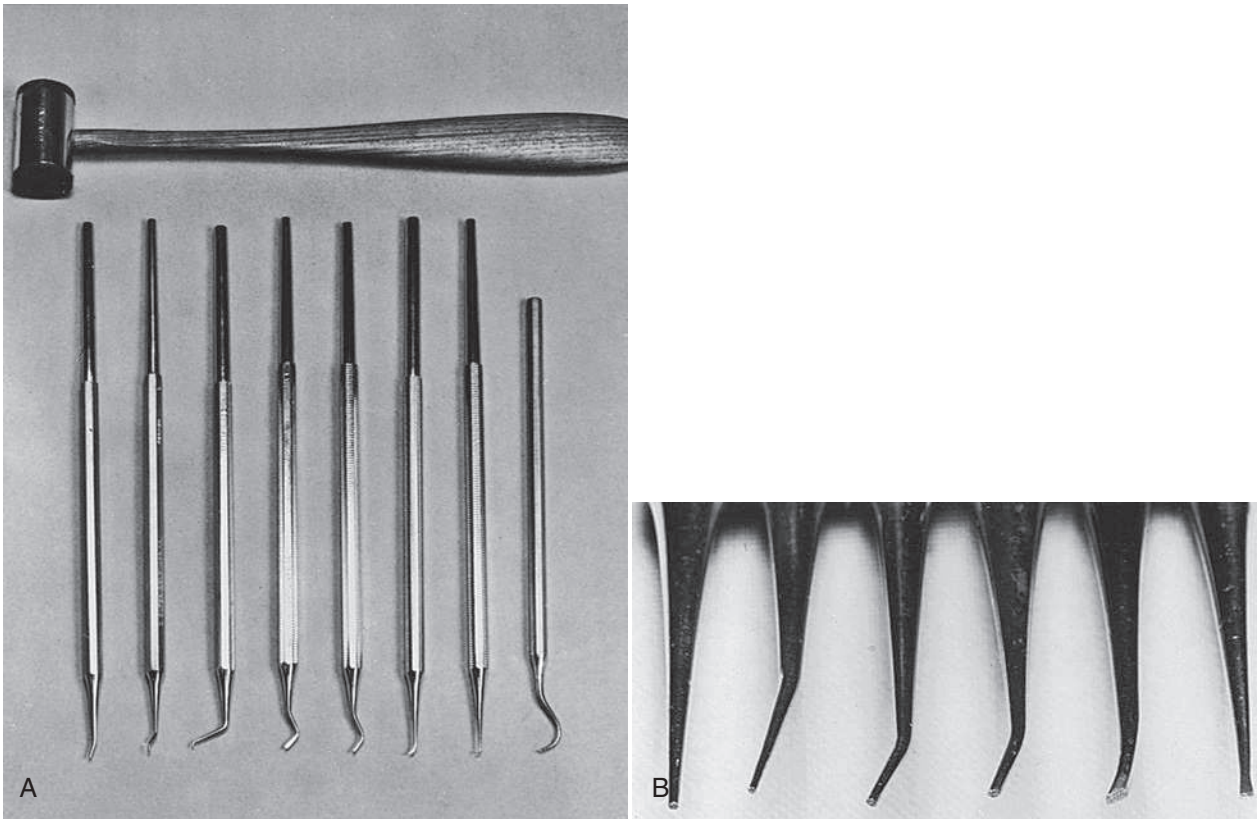
Two fundamental principles involved in compaction of cohesive gold are to (1) weld the gold into a cohesive mass and (2) wedge as much gold as possible into the tooth preparation.[15] Welding



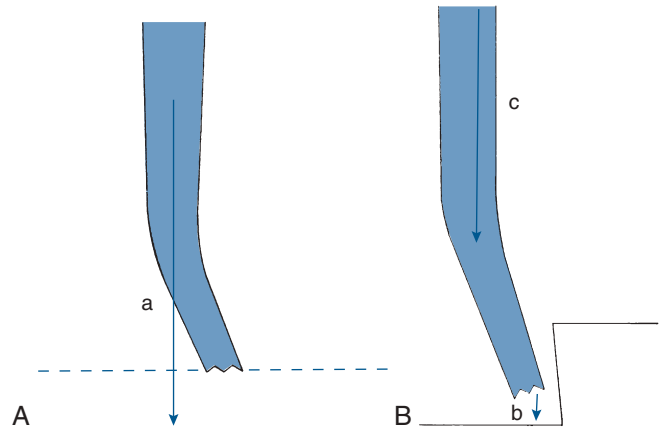
• **Fig. 17.3** Scanning electron micrographs of direct-filling golds. A, Spheres of E-Z Gold. B, Wrapped E-Z pellet that contains spheres. (Courtesy Ivoclar-Williams Company, Inc., Amherst, NY.)



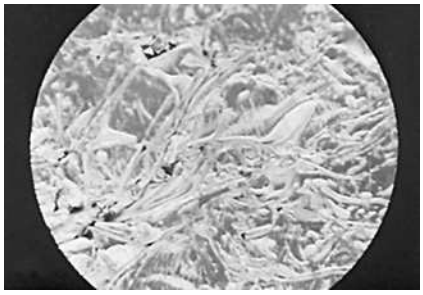
• **Fig. 17.4** A, Pellet of gold foil is degassed in pure ethanol flame. B, Mica tray mounted over alcohol lamp for degassing several increments of gold simultaneously. C, Gold foil degassed on an electric annealer. (Courtesy of Terkla and Cantwell.)



• **Fig. 17.5** A, Hand mallet and condensers used for hand mallet compaction of direct gold. B, Selection of variously shaped nibs. *Left to right*, Three round-faced nibs, oblique-faced nib, foot condenser, and rounded rectangular nib. (A, Courtesy of Terkla and Cantwell.)



• **Fig. 17.6** A, Oblique-faced condenser with the nib face established perpendicular to long axis of handle and perpendicular to line of force *(a)*. B, Conventional monangle condenser; the nib face is not perpendicular to line of force *(b)*; the condenser nib face is established perpendicular to end portion of shank rather than perpendicular to handle *(c)*.



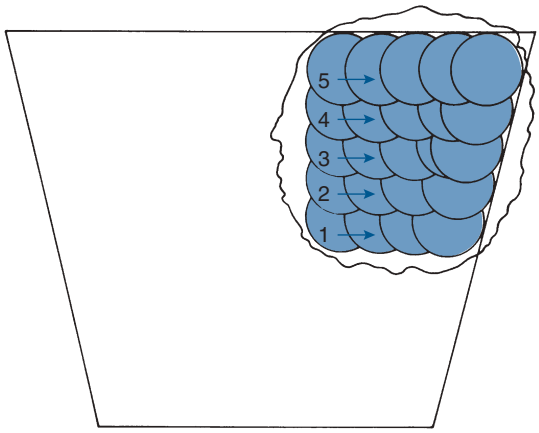
• **Fig. 17.7** Compacted gold foil. Linear channels are evident between creases in the foil pellet. Dark spots are void spaces in the compacted mass.

takes place primarily as a result of the coherence of the noble metal to itself. Wedging results from careful compacting technique. Regardless of the technique used, some bridging occurs, resulting in void spaces not only in the compacted gold but also along the preparation walls. Success depends on minimizing these voids, particularly on the surface of the restoration and at the cavosurface interface, where leakage to the internal aspects of the restoration may begin. Gold foil compacts readily because of its thin form and produces a mass with isolated linear channels of microporosity (Fig. 17.7). Because the thin folds of the gold pellet weld to each other, the remaining channels of microporosity do not appear to be entirely confluent with one another.

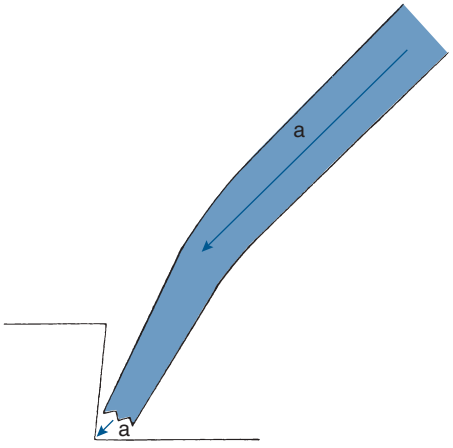
It is recommended that compaction of E-Z Gold be done by hand pressure. As compaction is performed, the bag of atomized gold is opened and the spheres of gold powder move over one another and against the preparation walls. Heavy and methodic hand pressure with the condensers is required to compact this form of gold effectively.

Compaction Technique for Gold Foil

Compaction begins when a piece of gold is placed in a tooth preparation. The gold is first pressed into place by hand, then a condenser of suitable size is used to begin malleting in the center of the mass (often this is done while this first increment is held in position with a holding instrument). Each succeeding step of



• **Fig. 17.8** Diagrammatic order of compaction for increment of direct-filling gold. Condensers are moved across surface of gold in an orderly stepping motion. Each succeeding step of the nib overlaps the previous one by at least half of the nib face diameter. Condensation begins at position 1 and moves to the right, then resumes at 2 and repeats movement to the right. Finally, it continues in rows 3, 4, and 5.



• **Fig. 17.9** Line of force *(a)* remains parallel with the shaft or handle of the condenser, regardless of any angles in the shank of the instrument.

the condenser overlaps (by half) the previous one as the condenser is moved toward the periphery (Fig. 17.8). The gold moves under the nib face of the condenser, effecting compaction as malleting proceeds.

The most efficient compaction occurs directly under the nib face.[15] Some compaction also occurs by lateral movement of the gold against surrounding preparation walls. The result of compaction is to remove most of the void space from within each increment of gold, to compact the gold into line and point angles and against walls, and to attach it to any previously placed gold via the process of cohesion.[16]

The line of force is important when any gold is compacted. The line of force is the direction through which the force is delivered (i.e., the direction in which the condenser is aimed) (Fig. 17.9). Specific instructions regarding line of force are given in subsequent sections as they relate to the restorations.

Research has shown that a biologically acceptable pulpal response occurs after proper direct gold procedures.[17] Care is required when condensing forces are applied to preclude pulpal irritation. The Electro-Mallet is an acceptable condenser if the manufacturer's

instructions for mallet intensity are followed. Correct hand-malleting technique requires a light, bouncing application of the mallet to the condenser, rather than delivery of heavy blows.

Compaction Technique for E-Z Gold

Using an amalgam condenser or a gold foil condenser, the first pellet of E-Z Gold is pressed into the depth of the tooth preparation and tamped into position. A small condenser is selected to thrust and wedge the gold into opposing line angles and against opposing walls, to secure the mass in the preparation. Additional pellets are added (one at a time, banking against the preparation walls) until the entire preparation is filled. To avoid creation of large void spaces in the restoration, a dense, fully condensed surface is obtained with each pellet before subsequent pellets are added.

Principles of Tooth Preparation for Direct Gold Restorations

Fundamentals of Tooth Preparation

The principles of tooth preparation for all direct gold restorations demand meticulous attention to detail for success. Failure to give attention to outline form may result in an unsightly restoration or, at the least, one in which cavosurface deficiencies are immediately obvious. Poor resistance form can result in tooth fracture; inadequate retention form may result in a loose restoration that is frustrating to the dentist. Lack of detailed convenience form may render an otherwise excellent tooth preparation unrestorable. The preparation must be smoothed and debrided to permit the first increments of gold to be stabilized.

The margins in outline form must not be ragged. They are established on sound areas of the tooth that can be finished and polished. The outline must include all structural defects associated with the lesion. The marginal outline must be designed to be esthetically pleasing because the final restoration may be visible.

Resistance form is established by orienting preparation walls to support the integrity of the tooth, such as a pulpal wall that is flat and perpendicular to occlusal forces. All enamel must be supported by sound dentin. Optimally placed axial or pulpal walls promote the integrity of the restored tooth, providing a suitable thickness of remaining dentin.

The retention form is established by parallelism of some walls and by strategically placed converging walls (as described in detail for each tooth preparation). In addition, walls must be smooth and flat where possible (to provide resistance to loosening of the gold during compaction), and internal line angles must be sharp (to resist movement). Internal form includes an initial depth into dentin, ranging from 0.5 mm from the dentinoenamel junction (DEJ) in Class I preparations to 0.75 mm from cementum in Class V preparations.

Optimal convenience form requires suitable access and a dry field provided by the rubber dam. Access additionally may require the use of a gingival retractor for Class V restorations or a separator to provide a minimal amount of separation (0.5 mm maximum) between anterior teeth for Class III restorations. Sharp internal line and point angles are created in dentin to allow convenient "starting" gold foil as compaction begins. Rounded form is permitted when E-Z Gold is used to begin the restorative phase. Removal of remaining carious dentin, final planing of cavosurface margins, and debridement complete the tooth preparation for direct gold.

Indications and Contraindications

Class I direct gold restorations are one option for the treatment of small carious lesions in pits and fissures of most posterior teeth and the lingual surfaces of anterior teeth. Direct gold also is indicated for treatment of small, cavitated Class V carious lesions or for the restoration, when indicated, of abraded, eroded, or abfraction areas on the facial surfaces of teeth (although access to the molars is a limiting factor). Class III direct gold restorations can be used on the proximal surfaces of anterior teeth where the lesions are small enough to be treated with esthetically pleasing results. Class II direct gold restorations are an option for restoration of small cavitated proximal surface carious lesions in posterior teeth in which marginal ridges are not subjected to heavy occlusal forces (e.g., the mesial or distal surfaces of mandibular first premolars and the mesial surface of some maxillary premolars). Class VI direct gold restorations may be used on the incisal edges or cusp tips. A defective margin of an otherwise acceptable cast-gold restoration also may be repaired with direct gold.

Direct gold restorations are contraindicated in some patients whose teeth have very large pulp chambers, in patients with severely periodontally weakened teeth with questionable prognosis, in patients for whom economics is a severely limiting factor, and in handicapped patients who are unable to sit for the long dental appointments required for this procedure. Root canal–filled teeth are generally not restored with direct gold because these teeth are brittle, although in some cases gold may be the material of choice to close access preparations (for root canal therapy) in cast-gold restorations.

Tooth Preparations and Restorations

This section presents the preparation and the restoration of Class I, V, and III lesions. The preparations described may be restored entirely with pellets of gold foil, or E-Z Gold may be used. If powdered gold is selected, heavy hand pressure compaction may be substituted for hand mallet or automatic mallet techniques. Class I and V E-Z Gold restorations may be veneered with gold foil pellets, if desired. The Class III tooth preparation discussed in this chapter is recommended by Ferrier, and only pellets of gold foil are used for the restoration. All tooth preparations and restorative procedures are accomplished after a suitable field of operation has been achieved (usually by application of rubber dam).

Class I Tooth Preparation and Restoration

Tooth Preparation Design

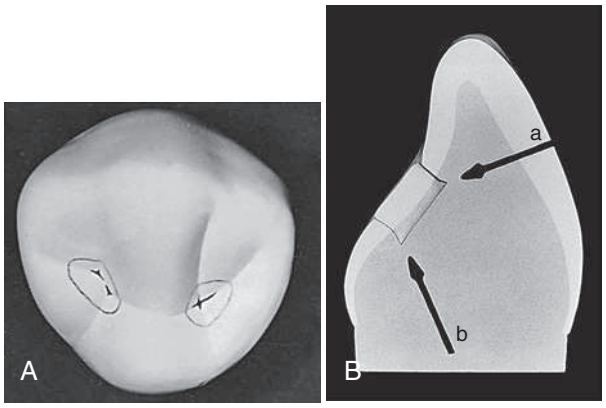
The marginal outline form for the Class I tooth preparation for compacted gold is extended to include the lesion on the tooth surface treated and any fissured enamel. The preparation outline may be a simple circular design for a pit defect or it may be oblong, triangular, or a more extensive form (if needed to treat a defective fissure) (Fig. 17.10A). Preparation margins are placed beyond the extent of pits and fissures. All noncoalesced enamel and structural defects are removed; the outline is kept as small as possible, consistent with provision of suitable access for instrumentation and for manipulation of gold.

For Class I tooth preparations, the external walls of the preparation are parallel to each other. In extensive occlusal preparations, the mesial or distal wall (or both) may diverge slightly occlusally, however, to avoid undermining and weakening marginal ridges. The pulpal wall is of uniform depth, parallel with the plane of the

surface treated, and established at 0.5 mm into dentin. The pulpal wall meets the external walls at a slightly rounded angle created by the shape of the bur. Small undercuts may be placed in dentin if additional retentive features are required to provide convenience form in beginning the compaction of gold (see Fig. 17.10B). Undercuts, when desired, are placed facially and lingually in posterior teeth (or incisally and gingivally on the lingual surface of incisors) at the level of the ideal pulpal floor position. These undercut line angles must not undermine marginal ridges. A slight cavosurface bevel may be placed to (1) create a 30- to 40-degree metal margin for ease in finishing the gold and (2) remove remaining rough enamel. The bevel is not greater than 0.2 mm in width and is placed with a white rotary stone or suitable finishing bur.

Instrumentation

For description and illustration, the preparation of a carious pit on the mandibular first premolar is presented (Fig. 17.11A). By use of a high-speed handpiece with air-water spray, the No. 330 or No. 329 bur is aligned, and the outline form (which includes the limited initial depth) is established (see Fig. 17.11B). When the preparation is extensive because of the inclusion of fissured enamel, a small hoe ($6\frac{1}{2}$-$2\frac{1}{2}$-9) may be used to complete the desired degree of flatness of the pulpal wall. With a No. $33\frac{1}{2}$ bur at low speed, small retentive undercuts are prepared into the dentinal portion of the external walls at the initial pulpal wall depth; these also may be prepared using a $6\frac{1}{2}$-(90)-$2\frac{1}{2}$-9 angle-former chisel. Round burs of suitable size are used to remove any infected carious dentin that remains on the pulpal wall. The preparation is completed by finishing the cavosurface with an angle former, a small finishing bur (e.g., No. 7802), or a flame-shaped white stone (see Fig. 17.11C–E).

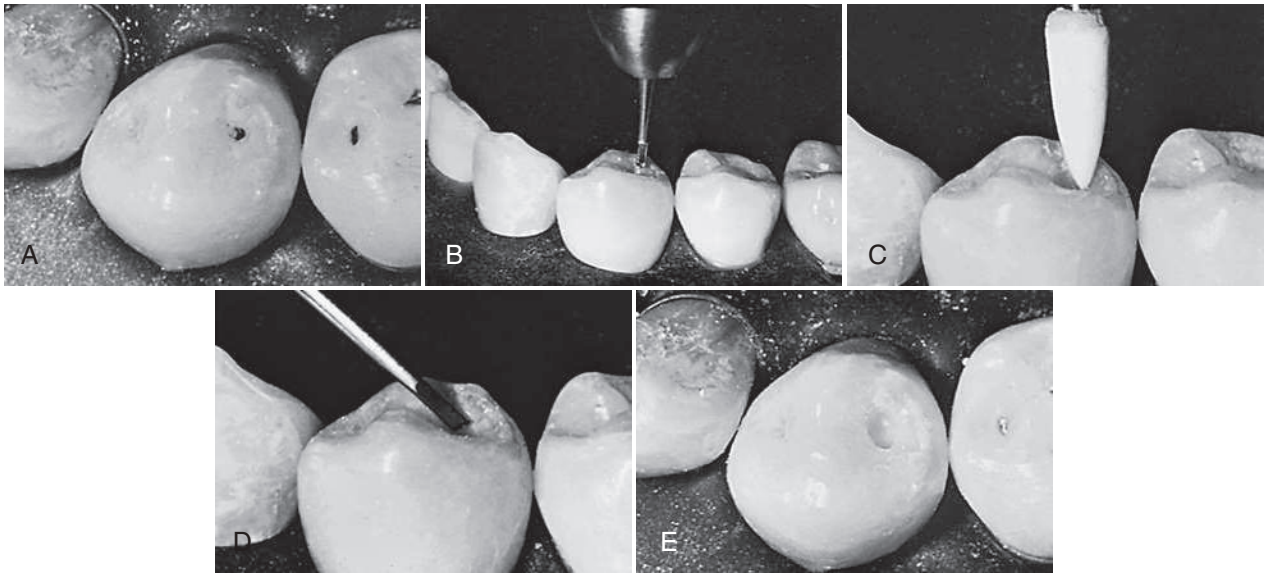


• **Fig. 17.10** A, Typical Class I occlusal marginal outlines for pit restorations with direct gold. B, Cross section of model of lingual Class I preparation on maxillary incisor. Undercuts *(a, b)* are placed in dentin incisally and gingivally for additional retention.

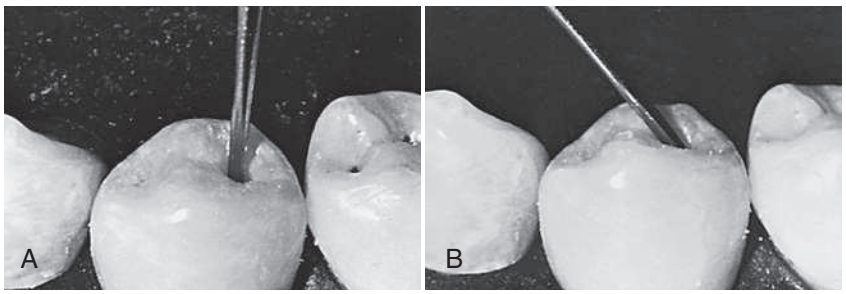
Restoration

The restorative phase begins with the insertion of a pellet of E-Z Gold or gold foil. The gold is first degassed in the alcohol flame, cooled in air for a few moments, and inserted into the preparation with the passing instrument. The gold is pressed into place with the nib of a small round condenser. In larger preparations a pair of condensers is used for this initial stabilization of the gold. Next, compaction of the gold begins with a line of force directed against the pulpal wall (Fig. 17.12A). Hand pressure is used for E-Z Gold; malleting is used for gold foil. The gold is compacted into the pulpal line angles and against the external walls, and the line of force is changed to a 45-degree angle to the pulpal and respective external walls (to compact the gold best against the internal walls) (see Fig. 17.12B). Additional increments of gold are added, and the procedure is repeated until the preparation is about three quarters full of compacted gold. If E-Z Gold is to be the final restoration surface, compaction is continued until the restoration is slightly overfilled.

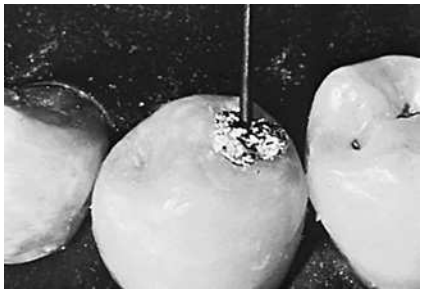
If gold foil is selected to veneer this restoration, pellets of suitable size are selected; in larger preparations, large pellets are convenient, whereas for small pit preparations, the operator should begin with $\frac{1}{64}$-size pellets (Fig. 17.13). The pellet is degassed and carried



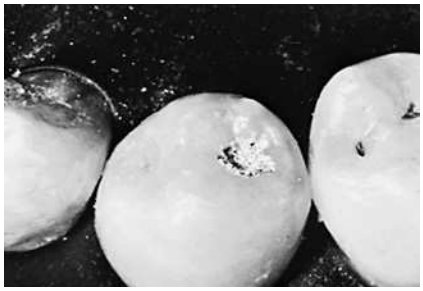
• **Fig. 17.11** Class I preparation for direct gold. A, Preoperative view of pit lesion. B, No. 330 bur is aligned properly for occlusal preparation. C, Occlusal cavosurface bevel is prepared with white stone. D, The bevel may be placed with an angle former. E, Completed tooth preparation.



• **Fig. 17.12** A, Compaction forces are delivered by the condenser held at 90-degree angle to the pulpal wall. B, Gold is condensed against the external preparation walls.



• **Fig. 17.13** Placement of pellet of gold foil and compaction into tooth preparation.



• **Fig. 17.14** Compaction of gold foil has proceeded sufficiently to cover all the cavosurface margins.

to the preparation. First, hand pressure compaction is used to secure the pellet against the compacted E-Z Gold and spread it over the surface; next, mallet compaction is used. Likewise, each succeeding pellet is hand compacted, and then is compacted with the mallet. The condenser point is systematically stepped over the gold twice as malleting proceeds. Generally the line of force is perpendicular to the pulpal floor in the center of the mass and at a 45-degree angle to the pulpal floor as the external walls are reached. At this stage and during all building of the restoration, the compacted surface should be saucer shaped, with the compaction of gold on the external walls slightly ahead of the center. The surface should never be convex in the center because this may result in voids in the gold and poor adaptation of the gold along the external walls when the condenser nib is "crowded out" along the wall by the center convexity. The operator continues building the restoration until the cavosurface margin is covered with foil (Fig. 17.14). One needs to exercise extreme care that gold is always present between the condenser face and the cavosurface margin; otherwise the condenser may injure (i.e., fracture) the enamel margin. The central area of the restoration's surface is filled in to the desired level. Tooth surface contour of the gold is created to simulate the final anatomic form, and a slight excess of gold is compacted on the surface to allow for the finishing and polishing procedures.

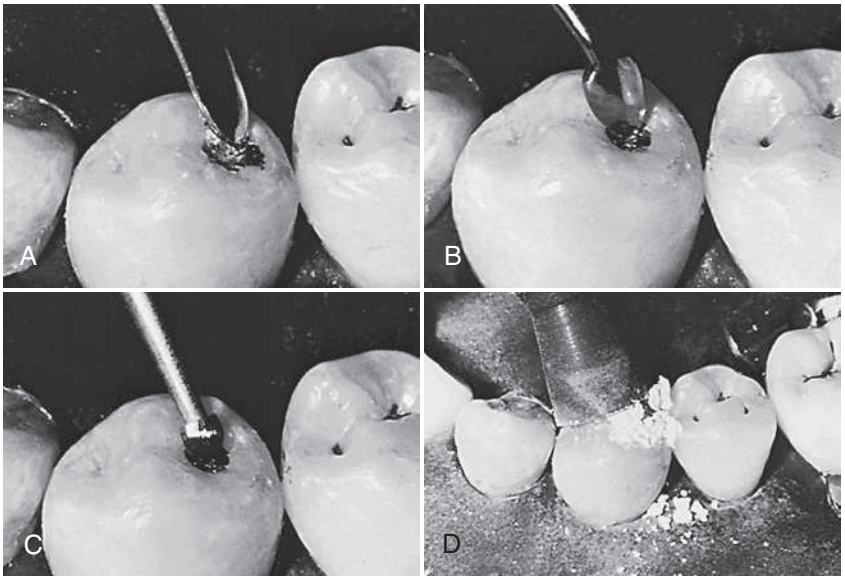
The first step in the finishing procedure is to burnish the gold (Fig. 17.15A). A flat beaver-tail burnisher is used with heavy hand pressure to harden the surface gold. A discoid-cleoid carver is used to continue the burnishing process and remove excess gold on the cavosurface margin. The cleoid, always directed so that a portion of the working edge is over or resting on enamel adjacent to or near the margins, is pulled from gold to tooth across the surface. This is done to smooth the surface and trim away excess gold (see Fig. 17.15B). If considerable excess gold has been compacted, a green stone may be necessary to remove the excess in Class I restorations. Care must be taken at this stage to avoid abrading the surface enamel. After use of the discoid-cleoid, a small round finishing bur (No. 9004) is used to begin polishing (see Fig. 17.15C). It is followed by the application of flour of pumice and tin oxide or white rouge (see Fig. 17.15D). These powdered abrasives are applied dry, with a webless, soft rubber cup in a low-speed handpiece. Care is taken to use light pressure. Gentle blasts of air cool the surface during polishing. The completed restoration is illustrated in Fig. 17.16.

Class V Tooth Preparation and Restoration

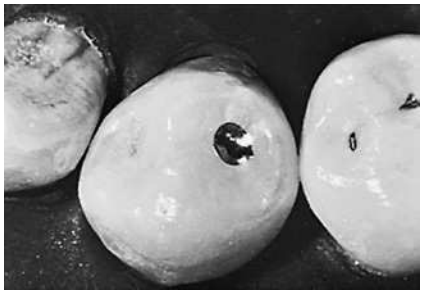
Operating Field

As with all direct gold restorations, the rubber dam must be in place to provide a suitable, dry field for a Class V restoration. For lesions near the gingiva or that extend into the gingival sulcus, it is necessary to provide appropriate access to the lesion by placing a No. 212 retainer or gingival retractor. The punching of the rubber dam is modified to provide ample rubber between teeth and to provide enough rubber for coverage and retraction of the soft tissue on the facial side of the tooth. The hole for the tooth to be treated is punched 1 mm facial of its normal position, and an extra 1 mm of dam is left between the hole for the treated tooth and the holes for the immediately adjacent teeth.

Several modifications may be made to the No. 212 retainer to facilitate its use. If the notches that are engaged by the retainer forceps are shallow, they may be deepened slightly with a large, carbide fissure bur to provide a more secure lock for the forceps (Fig. 17.17A). If the tips of the retainer jaws are very sharp, they may be slightly rounded with a garnet disk, then polished to avoid scratching cementum during placement. For application to narrow teeth (e.g., mandibular incisors), the facial and lingual jaws may be narrowed by grinding with a heatless stone or carborundum disk, after which they are polished with a rubber wheel. To expedite



• **Fig. 17.15** Steps in finishing Class I direct gold restoration. A, Burnisher work-hardens the surface gold. B, The discoid-cleoid instrument removes the excess gold from the cavosurface margins. C, A No. 9004 bur is used to begin the polishing phase. D, Polishing abrasives are applied with a rubber cup.



• **Fig. 17.16** Completed restoration.

placement on rotated teeth, the jaws may be modified by grinding suitable contour to the tip edge (see Fig. 17.17B). The jaws may be bent for use on teeth where gingival access to lesions is difficult. This is done by heating the jaws to a cherry-red color in a flame, then grasping the entire facial jaw with suitable pliers and slightly bending the jaw apically. The procedure is repeated for the lingual jaw, bending it slightly occlusally (Fig. 17.18).

The No. 212 retainer must be applied carefully to avoid damage to soft or hard tissue. The retainer is secured in the retainer forceps and carried to the mouth after the rubber dam has been placed. The lingual jaw is positioned just apical to the lingual height of contour, and the index finger is placed against the jaw to prevent its movement. The retainer is rotated faciogingivally with the forceps, while the thumb retracts the dam; the facial jaw is set against the tooth (Fig. 17.19A). Next, a ball burnisher is placed into one of the retainer notches and used to move the facial jaw gingivally (without scraping the jaw against the tooth) to the final position (i.e., 0.5–1 mm apical of the expected gingival margin) (see Fig. 17.19B). Gentle pressure is used to position the facial jaw so that only the free gingiva is retracted, and the epithelial attachment is not harmed. The retainer is supported and locked into this desired position with the compound, which is softened, molded by the fingers, and placed between the bows and the gingival embrasures (see Fig. 17.19C). The compound also serves to distribute compaction forces among all the teeth included in the retainer application.

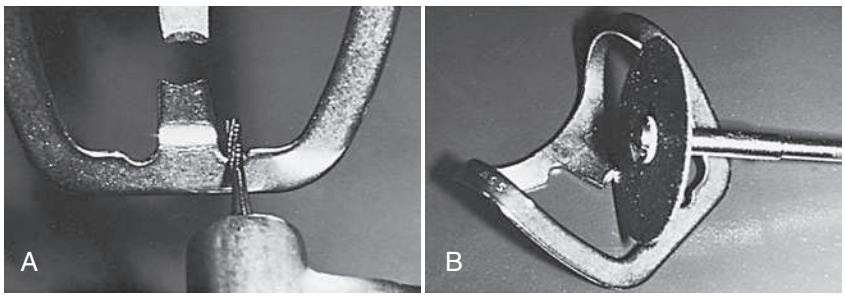
Tooth Preparation Design

The typical Class V tooth preparation for restoration with direct gold is trapezoidal (Figs. 17.20, 17.21, and 17.22). This outline form is created to satisfy esthetic needs and the requirements for the retention and convenience forms in the treatment of lesions in the gingival third of the clinical crowns of teeth. The straight occlusal margin improves the esthetic result, and by virtue of its straight design, excess gold is readily discerned and removed in the final stages of the restorative process. The gingival outline is shorter than the occlusal route because the tooth narrows in the gingival area. In addition, it is prepared parallel with the occlusal margin for easy identification in the finishing phases. The mesial and distal margins connect the gingival margin to the occlusal margin.

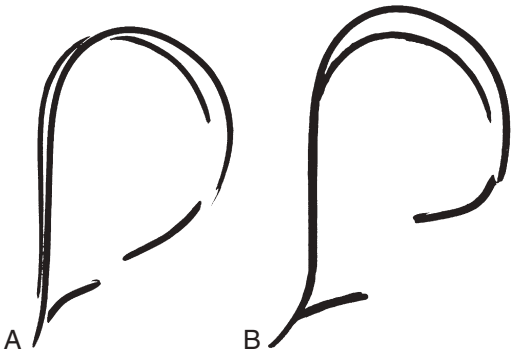
The occlusal margin is straight and parallel with the occlusal plane of the teeth in the arch (see Fig. 17.20); it is extended occlusally to include the lesion. (When several adjacent teeth are restored, some additional extension is permissible to create a uniform level that may be more esthetically pleasing.) Often, the mesiodistal extension to the line angles of the tooth places the junction of the occlusal and mesial and distal margins gingival to the crest of the free gingiva, rendering the most esthetic result. The gingival margin is also straight, parallel with the occlusal margin, placed only far enough apically to include the lesion, and it extends mesiodistally to the line angles of the tooth.

The mesial and distal margins are parallel to the proximal line angles of the tooth (see Fig. 17.22A) and usually are positioned sufficiently mesially and distally to be covered by the free gingiva. The mesial and distal margins are straight lines that meet the occlusal margin in sharp, acute angles and meet the gingival margin in sharp, obtuse angles, both of which complete the trapezoidal form.

The depth of the axial wall varies with the position of the preparation on the tooth. The axial wall is approximately 1 mm



• **Fig. 17.17** A, Notches are deepened for secure holding of the No. 212 retainer. B, Jaws may be modified with a disk to facilitate retainer placement on rotated teeth.



• **Fig. 17.18** A, Drawing of a No. 212 retainer as received from the manufacturer. B, Modified facial and lingual jaws.



• **Fig. 17.19** Placement of No. 212 retainer. A, Initial placement of facial jaw after first placing lingual jaw. B, Use of ball burnisher to carry the facial jaw to the final position. C, Retainer stabilized with compound to distribute compaction forces, prevent tipping, and to prevent either apical or occlusal movement of retainer.

deep in the occlusal half of the preparation. As the outline approaches the cervical line, the axial wall depth may decrease from 1 to 0.75 mm. The axial wall must be established in dentin, and occlusogingivally it should be relatively flat and parallel (approximately) with the facial surface of the tooth (see Fig. 17.22B). Mesiodistally, the axial wall also is prepared approximately parallel with the surface contour of the tooth. This contour may create a slight mesiodistal curvature in the axial wall in convex contoured teeth and where the preparation is extensive proximally. Mesiodistal curvature of the axial wall prevents encroachment of the tooth preparation on the pulp. Excessive axial curvature results in a preparation that is either too shallow in the center or too deep at the proximal extensions, and it further complicates restoration by failing to provide a reasonably flat wall against which to begin compaction. A subaxial wall may be created within the axial wall to remove infected caries that has progressed deeper than the ideal axial wall placement.

The occlusoaxial internal line angle is a sharp right angle. The occlusal wall also forms a right angle with the external enamel surface, precluding undermining of the enamel. The gingivoaxial internal line angle is a sharp, acute angle, created at the expense of the gingival wall (see Fig. 17.22B). The mesioaxial and distoaxial internal line angles are sharp, obtuse angles. These obtuse line angles are created to prevent the undermining of the mesial and distal enamel, although still providing some resistance to movement of the gold during compaction. They must never be acute angles.

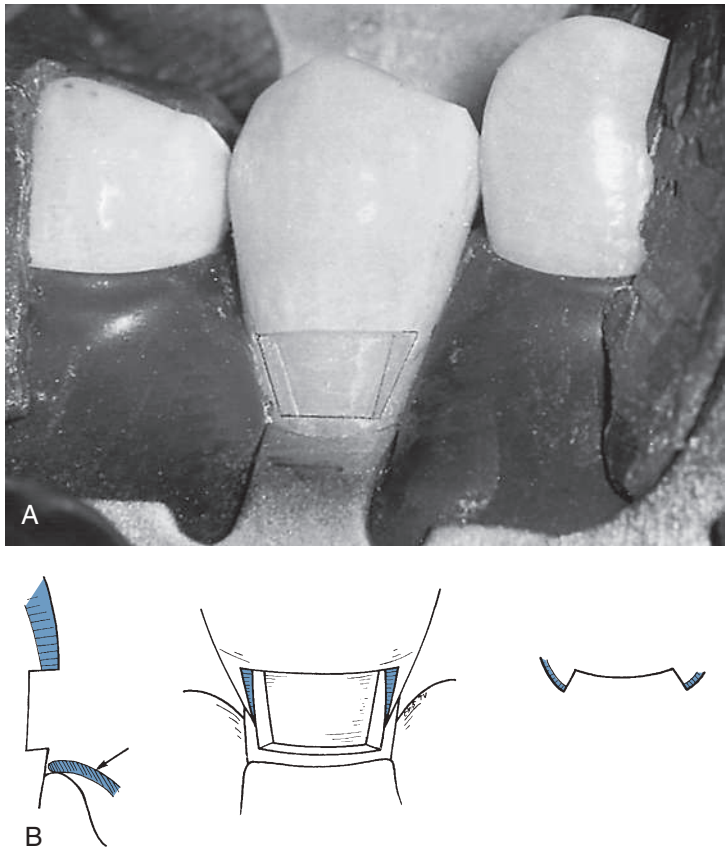
The mesial and distal prepared walls are flat and straight. They meet the occlusal wall in a sharp, acute line angle and meet the gingival wall in a sharp, obtuse line angle. The mesial and distal



• **Fig. 17.20** Facial view of Class V tooth preparation for direct gold. The occlusal and gingival margins are straight, parallel with each other, and extend mesially and distally to the respective mesiofacial and distofacial tooth crown line angles. The mesial and distal walls diverge facially and form obtuse angles with the axial wall. Line angles and point angles are sharp (see also Fig. 17.22B).



• **Fig. 17.21** Facioocclusal view of design of gingival wall in Class V preparation for direct gold. The axiogingival line angle is acute and has been prepared at the expense of the gingival wall. This gingival margin is on cementum. If on enamel, the gingival cavosurface would be beveled slightly (see also Fig. 17.26E).



• **Fig. 17.22** A, Clinical Class V tooth preparation. Note the proper isolation of the operating field. This gingival margin is on cementum. B, Longitudinal section, facioocclusal view, and cross section. Line and point angles are sharp.

walls provide resistance for gold compaction, but they provide no retention.

The orientation of the gingival wall is the key to the retention form of the preparation. It is straight mesiodistally, meeting the mesial and distal walls in sharp line angles. Retention is provided by sloping the gingival wall internally to meet the axial wall in a sharply defined acute line angle. Retention is provided by the facial convergence of the occlusal and gingival walls. Gold wedged between these two walls is locked into the tooth. If the gingival margin is established on enamel, the cavosurface is beveled slightly to remove

unsupported enamel (see Fig. 17.26E later in the chapter). When placed on cementum, the gingival cavosurface is not beveled (see Fig. 17.24B later in the chapter).

The outline of the preparation may be modified. In clinical situations demanding reduced display of gold, such as in anterior teeth, the incisal outline may be curved to follow the contour of soft tissue mesiodistally (Fig. 17.23). This modification is made only when required because preparation instrumentation and finishing of gold are more difficult than when a straight marginal outline is created. A similar modification may be made in the occlusal outline when caries extends more occlusally as the proximal extensions are reached. Also the mesiodistal extension (i.e., dimension) of a preparation may be limited when caries is minimal, conserving intact tooth structure. When access requires, the gingival wall may be modified also to curve mesiodistally to include the gingival extent of advanced caries. The entire axial wall should not be extended pulpally to the depth of the lesion when deep cervical abrasion, abfraction, or erosion is treated; rather the axial wall is positioned normally, leaving a remaining V notch at its center to be restored with gold. When failing restorations are removed and restored with direct gold, the preparation outline is partially dictated by the previous restoration (Fig. 17.24).

Instrumentation

The No. 33½ bur is used to establish the general outline form of the preparation. The end of the bur establishes the distal wall (Fig. 17.25A); the side establishes the axial depth and the occlusal, gingival, and mesial walls (see Fig. 17.25B). When access permits, the end of the bur may be used to establish the mesial and gingival walls (see Fig. 17.25C and D). The gingival and mesial walls may be prepared with the side of the bur if access so dictates (see Fig. 17.25E and F). The end of the bur is used to place the axial wall in dentin (see Fig. 17.25G).

The 6½-2½–9 hoe or the larger 10-4-8 hoe is useful for planing preparation walls, establishing sharp internal line angles (Fig. 17.26A), and finishing margins. The Wedelstaedt chisel is used to finish the occlusal cavosurface margin (see Fig. 17.26B) and may be used to plane the axial wall. The acute axiogingival angle is established with the 6½-2½-9 hoe, cutting from the cavosurface to the axial wall in a push-cut stroke (see Fig. 17.26C). The chips of dentin produced at the axiogingival angle may be removed with the tip of an explorer (see Fig. 17.26D) or the point of a 6½-(90)-2½-9 small angle former. Care must be taken not to gouge the axial wall. When its use is indicated, the gingival bevel is prepared with the Wedelstaedt chisel or a hoe (see Fig. 17.26E).



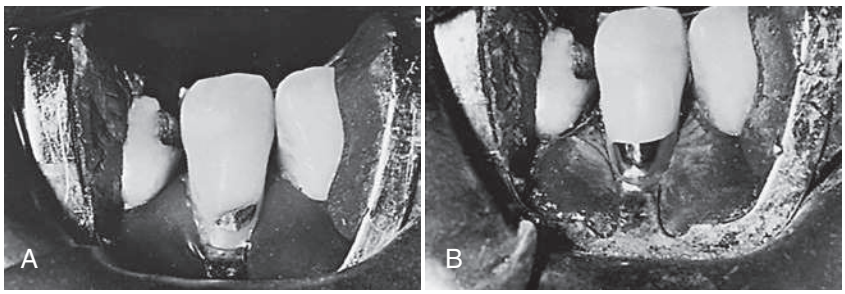
• **Fig. 17.23** Completed Class V gold restoration. Incisal margin curved to follow contour of gingival tissue for best esthetic result.

Restoration

Restoration of the Class V preparation begins with application of cavity varnish (if desired), after which a piece of degassed E-Z Gold is placed into the preparation. The gold is degassed in the alcohol flame and carried to its place in the preparation with the passing instrument. Parallelogram foil condensers or other suitable serrated condensers are used to force the gold firmly against the axial wall and to wedge it into the line angles. One instrument may be put aside (and the other is used as a holding instrument to prevent movement of the entire piece of gold), and compaction can begin by delivering heavy compacting forces to the gold.

After stabilization of the gold, completion of compaction of the initial mass of gold begins in the center of the mass with a 0.5-mm-diameter, round, serrated condenser nib. Careful, methodical stepping of the gold proceeds outward toward the external walls (to wedge the gold in the tooth and remove internal voids). As soon as the gold is stabilized, a holding instrument is no longer necessary. As the walls are reached, a line of force of 45 degrees to the axial wall is used to drive the gold into the line angles and against the external walls. The entire surface of the gold is condensed twice to complete the compaction of the gold. Additional increments of E-Z Gold are added until the preparation is filled to at least half its depth. E-Z Gold pellets are used to complete the restoration, covering the margins first, and to complete compacting in the center of the facial surface. Pellets of gold foil also may be used to complete the outer one half of the restoration (Fig. 17.27).

If gold foil is used for the outer half of the restoration, compaction proceeds with medium-sized pellets at the mesioocclusal or distoocclusal line angle and then across the occlusal wall. The entire wall and occlusal cavosurface margin are covered with compacted gold foil (see Fig. 17.27A). To ensure that gold protects

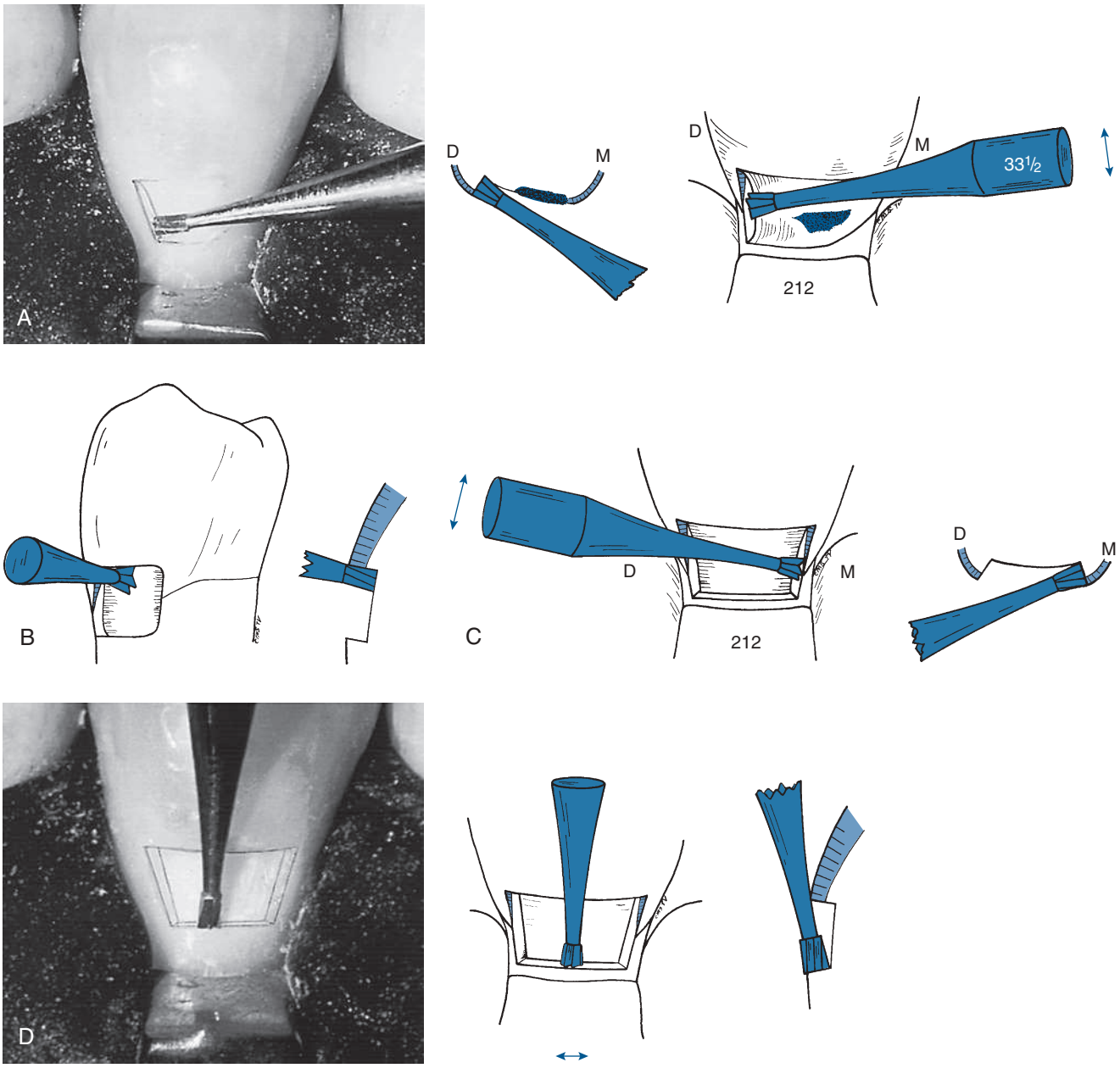


• **Fig. 17.24** A, Failing Class V amalgam restoration. B, Replacement direct gold restoration.

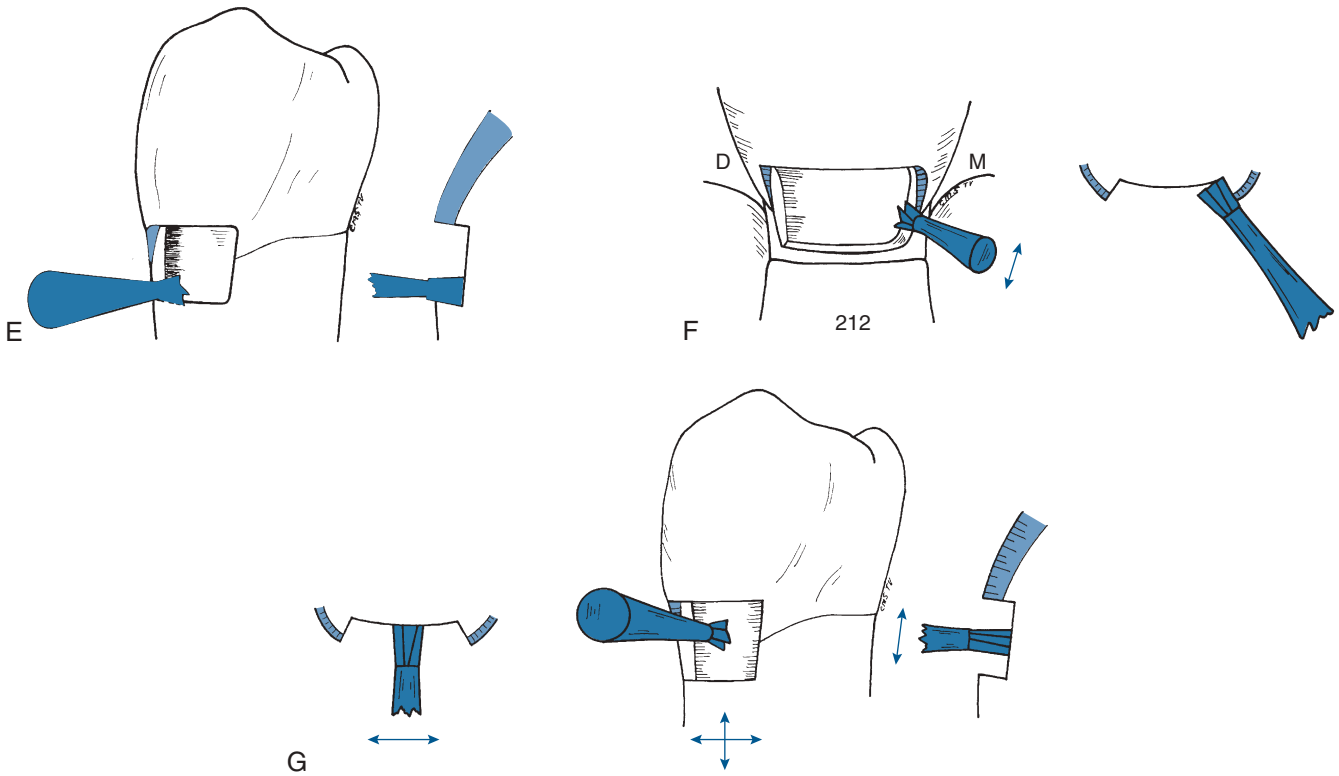
the margin from blows of the condenser face, care should be exercised when the condenser approaches any enamel margin. Next the gingival, mesial, and distal walls are covered, which leaves the restoration concave (see Fig. 17.27B). It is essential that all cavosurface margins be covered at this time, before the final convex surface of the restoration is formed.

Medium and large pellets (sizes ¼₃ and ½₂) are compacted in the center of the restoration to complete the formation of the appropriate contour. A slight excess contour is developed and is removed later when the gold is finished and polished. Any small remaining deficiencies in the surface contour are filled with small pellets. A Varney foot condenser (or other large condenser) is malleted over the entire surface to make it smooth and assist in detection of any poorly compacted areas (see Fig. 17.27C).

Finishing begins with application of a beaver-tail burnisher to work-harden and smooth the surface (Fig. 17.28A). Petroleum jelly may be applied to the dam to avoid abrasion from disks; it also may be applied to the disks. Gross excess contour, if any, is removed with a fine garnet disk applied with a Sproule or other suitable mandrel in a low-speed handpiece (see Fig. 17.28B). Excess gold is removed from the cavosurface margins with the discoid-cleoid instrument (using pull-cut strokes) or the gold knife (using only push-and-cut strokes from the gold to the tooth) (see Fig. 17.28C and D). When removing the excess gold over the gingival margin,



• **Fig. 17.25** Use of No. 33½ bur in straight handpiece for initiating Class V preparation. A, The end of the bur is used to establish the distal wall. B, The side of the bur is used to establish the occlusal wall. C, The end of the bur prepares the mesial wall, if access permits. D, The end of the bur is used to establish the gingival wall, if access permits. The use of a No. 33½ bur in the straight handpiece for initiating Class V preparation.



• **Fig. 17.25, cont'd** E, Preparation of the gingival wall with the side of the bur. F, Preparation of the mesial wall with the side of the bur. G, The end of the bur may be used to establish the initial axial wall depth in dentin.

care is exercised not to remove cementum or "ditch" the root surface (especially when using rotary instruments).

When the final contour has been obtained, cuttle disks may be used in decreasing abrasiveness (i.e., coarse to medium to fine) to ready the surface for final polishing. These disks and the cleoid are helpful in removing very fine fins of gold from margins. Polishing is performed with fine pumice followed by tin oxide or white rouge (applied with a soft, webless rubber cup). Care also is required at this stage to avoid ditching cementum with the polishing abrasive. The abrasives are used dry so that the field may be kept clean, and the exact position of the rubber cup can be seen at all times (Fig. 17.29).

After polishing has been completed, the No. 212 retainer and rubber dam are removed. Removal of the retainer is best accomplished with the forceps firmly locked into the notches on the retainer. The retainer jaws are opened from the tooth with the forceps and carefully removed occlusally (without scratching the restoration or the surface enamel of the tooth). The gingival sulcus is rinsed and examined to ascertain that it is free of debris. Soft tissue is massaged gently before the patient is dismissed.

Class III Tooth Preparation and Restoration

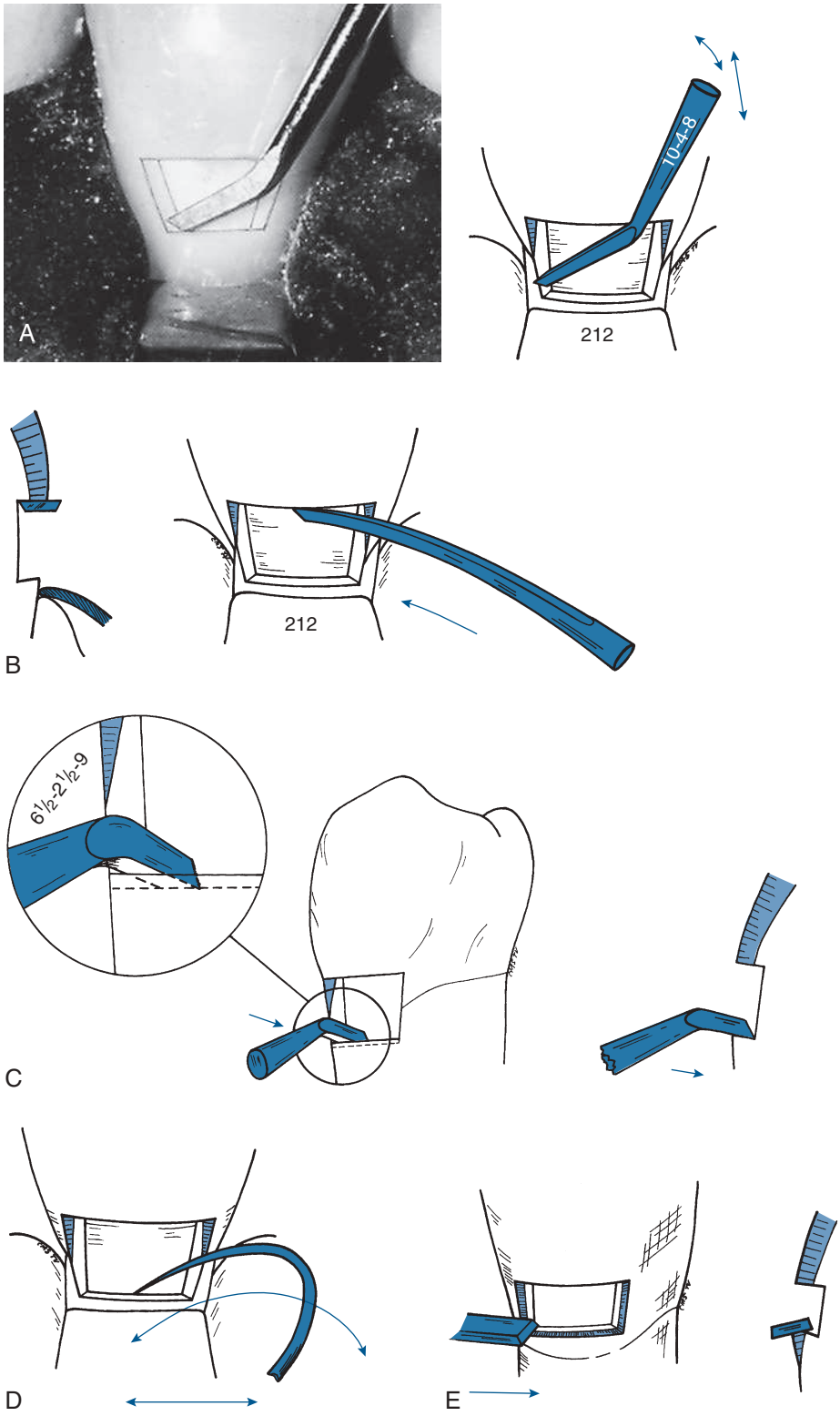
Many styles of Class III preparations are advocated for use with direct gold. Some preparations are based on the lingual approach and are restored with E-Z Gold. Others may be instrumented from either the facial or the lingual surface and use gold foil as the restorative material. The outline form selected must provide adequate access for placing the restoration and developing an acceptable esthetic result. The preparation design presented in subsequent sections was first described by Ferrier in the early years of the 20th century and is still used today.[18] It has the advantage of not only conserving the tooth structure but also providing access for compaction of gold foil directly against all preparation walls and cavosurface margins. This results in a dense, esthetically pleasing result (if careful attention is given to management of the outline design). This preparation is instrumented primarily from a facial approach, although lingual instrumentation may be used in maxillary teeth. The preparation may be modified for mandibular anterior teeth, the distal surface of maxillary canines, and the distal surface of some lateral incisors.

Tooth Preparation Design for Maxillary Incisors

The marginal outline is the most important. From a facial view, the gingival four fifths of the facial margin is straight and (generally) parallel with the contour of the tooth (Fig. 17.30). The facial margin forms a gentle curve in its incisal one fifth to blend with the incisal margin. When viewed from a proximofacial aspect, the facial outline follows the general contour of the adjacent tooth (Fig. 17.31) and meets the gingival outline in a slightly obtuse angle. This juncture may be curved slightly to enhance esthetics.

The gingival margin is crucial to the entire preparation. Its faciolingual length dictates the remainder of the preparation. Where possible, the gingival margin is established just apical to the crest of the free gingiva to enhance the esthetic result. It is straight faciolingually and is approximately at a right angle to the long axis of the tooth. It meets the facial margin in a sharply defined obtuse angle that may be rounded slightly (as previously described), and it meets the lingual margin in a sharply defined acute angle.

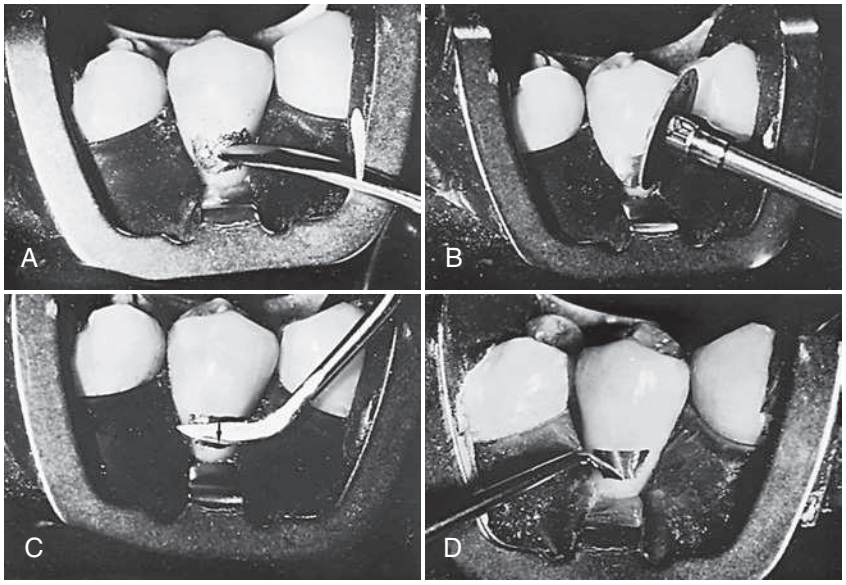
Viewed from the lingual aspect, the lingual margin generally parallels the long axis of the tooth (Fig. 17.32). It may diverge slightly proximally from the long axis, however, to parallel more



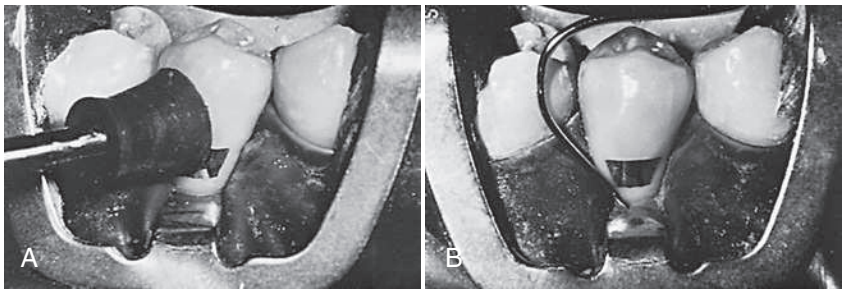
• **Fig. 17.26** Use of hand instruments in Class V tooth preparation. A, The small hoe planes the preparation walls. B, The Wedelstaedt chisel refines the occlusal wall and the margin. C, The small hoe creates an acute axiogingival line angle in dentin. D, The explorer is used to remove debris from the completed preparation. E, The chisel blade bevels the gingival cavosurface margin, when indicated. (E, From Howard WC, Moller RC: *Atlas of operative dentistry*, St. Louis, 1981, Mosby.)



• **Fig. 17.27** Completion of compaction where gold foil is used to overlay the E-Z Gold. A, Condensation of foil proceeds to cover the cavosurface margins. A slight excess of gold has been condensed over the mesial half of the occlusal cavosurface margin. B, All cavosurface margins are covered with a slight excess of gold. The restoration, at this stage of insertion, is concave. C, After additional foil pellets are compacted in the central area to form a convex restoration surface with slight excess, a foot condenser is used to confirm condensation.



• **Fig. 17.28** Finishing the Class V restoration. A, Burnisher work-hardens surface. B, A small, fine garnet disk removes the excess gold contour. C, The gold knife's secondary edge used with push-stroke *(arrow)* removes excess gold from the gingival margin. D, After final surfacing with a cuttle disk, any remaining marginal excess is removed with the cleoid carver.

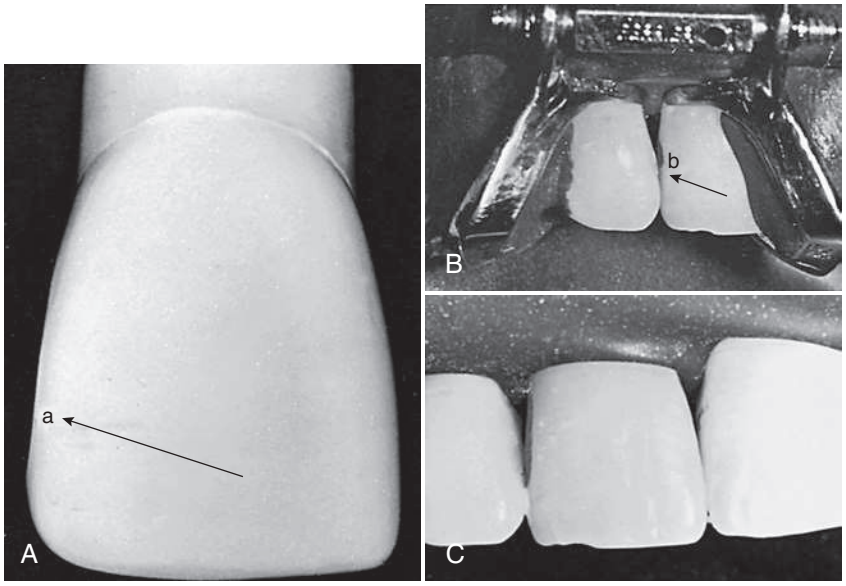


• **Fig. 17.29** A, A soft-rubber cup is used to apply polishing abrasives. B, The explorer is used to remove any remaining polishing powder from site of completed restoration.

nearly the proximal contour. It meets the gingival margin in a sharply defined angle that is nearly 90 degrees when viewed from the lingual aspect (Fig. 17.33), but it is acute when viewed from the proximal aspect. The lingual margin is straight in its gingival two thirds, but then it curves abruptly to meet the incisal margin.

The incisal margin is placed incisally to the contact area to provide access to the preparation; however, it is not extended enough to weaken the incisal angle of the tooth. It forms a smooth curve that connects the facial and lingual margins of the preparation.

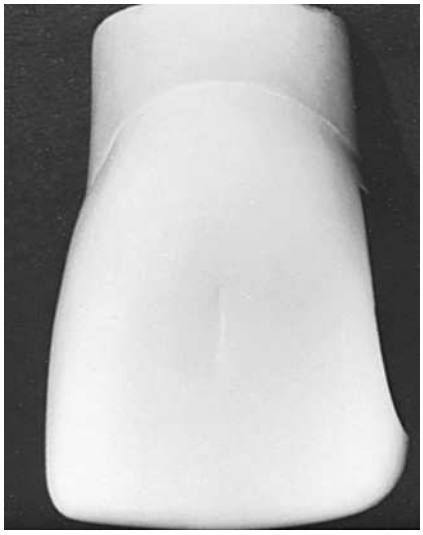
To provide a suitable resistance form, the internal aspects of the preparation are precisely instrumented. The gingival wall



• **Fig. 17.30** Class III direct gold restoration. A, The model of the preparation shows the esthetic marginal outline *(a)*. B, Central incisor *(b)* before distal preparation. C, Completed Class III restoration.



• **Fig. 17.31** Proximofacial view of Class III preparation.



• **Fig. 17.32** Lingual view of Class III preparation.

is flat faciolingually. The axial wall is flat faciolingually and incisogingivally, and it is established 0.5 mm into dentin. The resistance form also is created by establishing sharp, obtuse axiofacial and axiolingual line angles in dentin. The facial and lingual walls diverge only enough to remove undermined enamel, and yet they provide firm, flat walls against which the gold can be compacted.

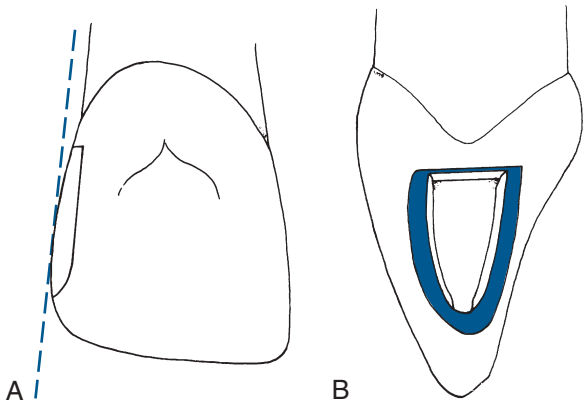
As in the Class V restoration, retention form is provided only between the gingival and incisal walls. In the Class III preparation, the dentinal portion of the gingival wall (as in the Class V gingival wall) slopes apically inward to create an acute axiogingival line angle. In the Class III preparation, the incisal portion is undercut (Fig. 17.34). This undercut is placed in dentin, facioincisally, to create a mechanical lock between the incisal and gingival walls. This increased retention form in the Class III preparation is required because of the length of the preparation incisogingivally and because of the difficulty of access in compacting the gold.

Provision for the convenience form is made by the abrupt incisolingual curve (which permits introduction of a condenser directed toward the gingival wall), by adequate clearance of all margins from the adjacent tooth, and by placement of sharp internal point angles suitable for beginning compaction of gold. The facioaxiogingival and linguoaxiogingival point angles may be enlarged slightly to assist in initial stages of foil compaction, if desired.[19]

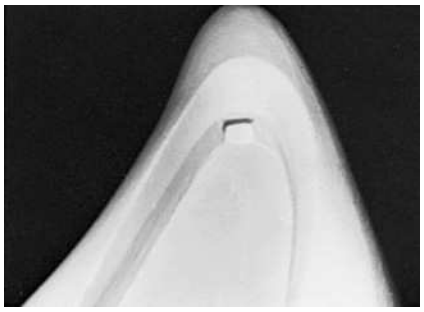
The finishing of enamel walls requires placement of a facioincisolingual cavosurface bevel, which determines the final marginal outline. This bevel is made with hand instruments and is established totally in enamel. It is designed to create maximum convenience form, to remove all surface irregularities and any unsupported enamel, and to establish a more esthetically pleasing result (Fig. 17.35; see also Fig. 17.30).

Modifications of Class III Preparations

The distal surface of maxillary canines may require a modification in preparation design for convenience in gold compaction. Because a highly convex surface is generally present, it is often desirable to create a "straight-line preparation" in which the facial outline appears as a slice. This modification provides clearance from the mesial marginal ridge of the first premolar and provides considerable convenience form to allow compaction of gold on the gingival wall directly from an incisal position. This type of preparation also is appropriate for the distal surface of highly contoured lateral incisors (Fig. 17.36).



• **Fig. 17.33** Lingual marginal outline of Class III preparation. A, View of lingual outline. Note the sharp linguogingival angle. B, Proximal view of preparation. Note that the linguogingival angle is sharp and acute in this view. (A, From Stibbs GD: Direct golds in dental restorative therapy. *Oper Dent* 5:107, 1980.)



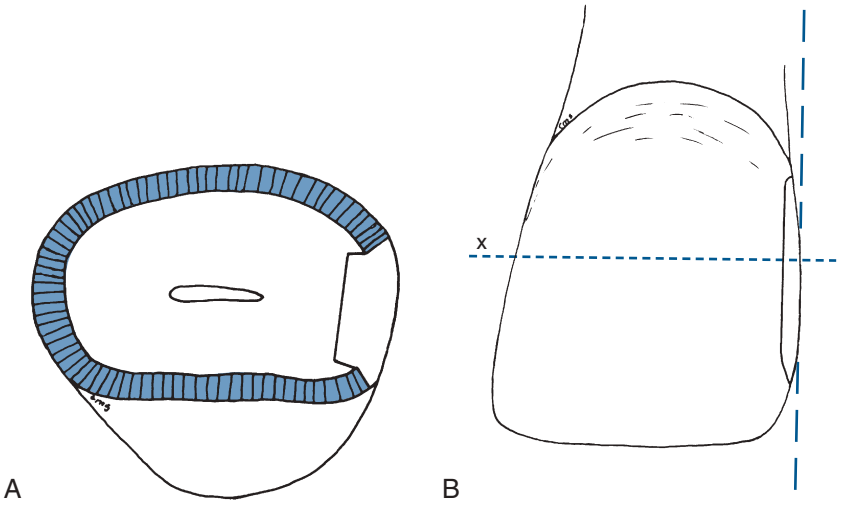
• **Fig. 17.34** View of incisal retention in Class III preparation. The undercut is placed in dentin but does not undermine enamel.

The mandibular incisors require a modified Class III preparation because of their small size and because access from a lingual position may be exceptionally difficult. The lingual wall is created in one plane, and extension of the lingual and the incisal walls is limited. The axiolingual line angle is a right or slightly obtuse angle. Care is taken to avoid lingual overextension of the lingual wall because this can result in the removal of dentinal support for lingual enamel, rendering the preparation unrestorable by direct gold. The outline form is extended lingually only far enough to include the lesion and to allow access for finishing of the gold. Incisal extension is restricted because the proximal contact area between mandibular incisors is often near the incisal angle. Extension incisally past the contact may weaken this critical area of the tooth; a mechanical separator may be necessary to obtain clearance between teeth. This provides access for tooth preparation and gold compaction. Facial extension is similar to the maxillary preparation (Fig. 17.37).

Internally, the incisal retentive angle for the mandibular Class III preparation is placed directly incisally, rather than facioincisally as in maxillary teeth. This modification is made to conserve the thickness of the tooth structure at the facioincisal angle, where wear of mandibular anterior teeth frequently occurs. Lingual approach Class III restorations may be made using E-Z Gold. In such cases, the lingual "slot" type of preparation is made with rounded internal line angles.

Separation of Teeth

Separation of teeth frequently is needed for instrumentation or finishing procedures performed on Class III direct gold restorations. The Ferrier separator is a convenient instrument for accomplishing this separation. It is applied and stabilized with compound (similar to stabilization of a No. 212 retainer) (Fig. 17.38). The jackscrews of the separator are activated with the separator wrench to draw the teeth slightly apart, creating a maximum space of 0.25 to



• **Fig. 17.35** Class III preparation internal form and facial marginal outline. A, Incisal view of cross section of preparation in plane *x* shown in *B*. Facial and lingual cavosurface bevels are shown placed in enamel. B, Facial view of the facial marginal outline of the preparation. (From Stibbs GD: Direct golds in dental restorative therapy. *Oper Dent* 5:107, 1980.)

0.5 mm. It is desirable to provide only this minimum separation and to remove the separator as soon as possible (preventing damage to periodontal structures).

Instrumentation

The No. 33½ bur (or a suitable Wedelstaedt chisel) is used to begin the preparation (Fig. 17.39). The bur is angled from the facial to position the gingival outline and the facial wall. A Wedelstaedt chisel establishes the lingual extension, and the No. 33½ bur defines the linguogingival line angle (Fig. 17.40) and completes the gingival floor preparation. The outline form is completed by beveling the cavosurface areas with a Wedelstaedt chisel. Next the dentinal part of the gingival, lingual, facial, and incisal walls is planed. A small hoe (i.e., 6½-2½-9) is used for the lingual and gingival walls (Fig. 17.41). An angle former is used to plane the facial dentinal wall (Fig. 17.42). An axial plane (i.e., 8-1-23) smooths the axial wall, and a bibeveled hatchet (i.e., 3-2-28) establishes the incisal retentive angle with a chopping motion (Fig. 17.43). Small angle formers are used to complete the sharp facioaxiogingival and linguoaxiogingival point angles and the slightly acute axiogingival angle (Fig. 17.44). The point angles may be enlarged further with the No. 33S bur (i.e., end-cutting bur) for additional convenience form. The Wedelstaedt chisel may be used again to complete the final planing of the cavosurface margins (Fig. 17.45).

Restoration

The separator is used to obtain a separation of 0.25 to 0.5 mm. Compaction of gold foil begins at the linguoaxiogingival point angle (Fig. 17.46). A small (i.e., 0.4 mm) monangle condenser is used to compact the gold, which is held by a small holding instrument. Pellets size ¼64 or ¼128 are used in the beginning of the restorative phase. The line of force is directed over the facial surface of the adjacent tooth and into the linguoaxiogingival point angle (see Fig. 17.46B). As soon as ample gold has been compacted into the linguogingival area to cover the linguogingival shoulder, compaction continues across the gingival wall (Fig. 17.47) and into the faciogingival angle. The offset condenser (with a faciogingival line of force) is used to fill the facioaxiogingival point angle (Fig. 17.48). Compaction of gold at the linguogingival area is confirmed with the oblique-faced monangle condenser (i.e., 0.5 mm) from the linguoincisal position (Fig. 17.49). Failure to provide dense gold in this linguogingival area at this stage may result in a void at the linguogingival angle and subsequently may lead to restoration failure.

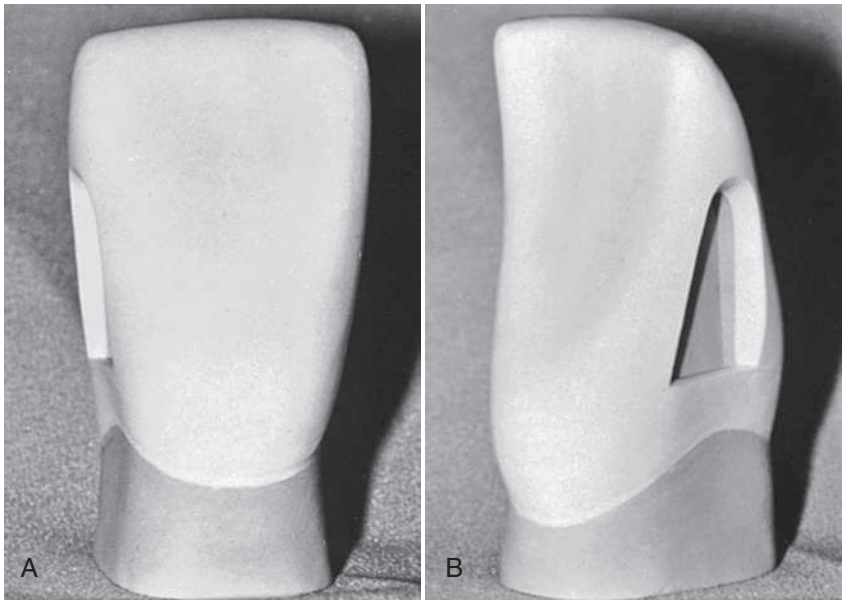
The bulk of the restoration is compacted with ¼43- or ¼32-sized pellets, mainly from a facial (occasionally from a lingual) direction (Fig. 17.50). The line of force is maintained in an axiogingival direction with the 0.5-mm monangle or oblique-faced monangle condenser (see Fig. 17.50B). This requires



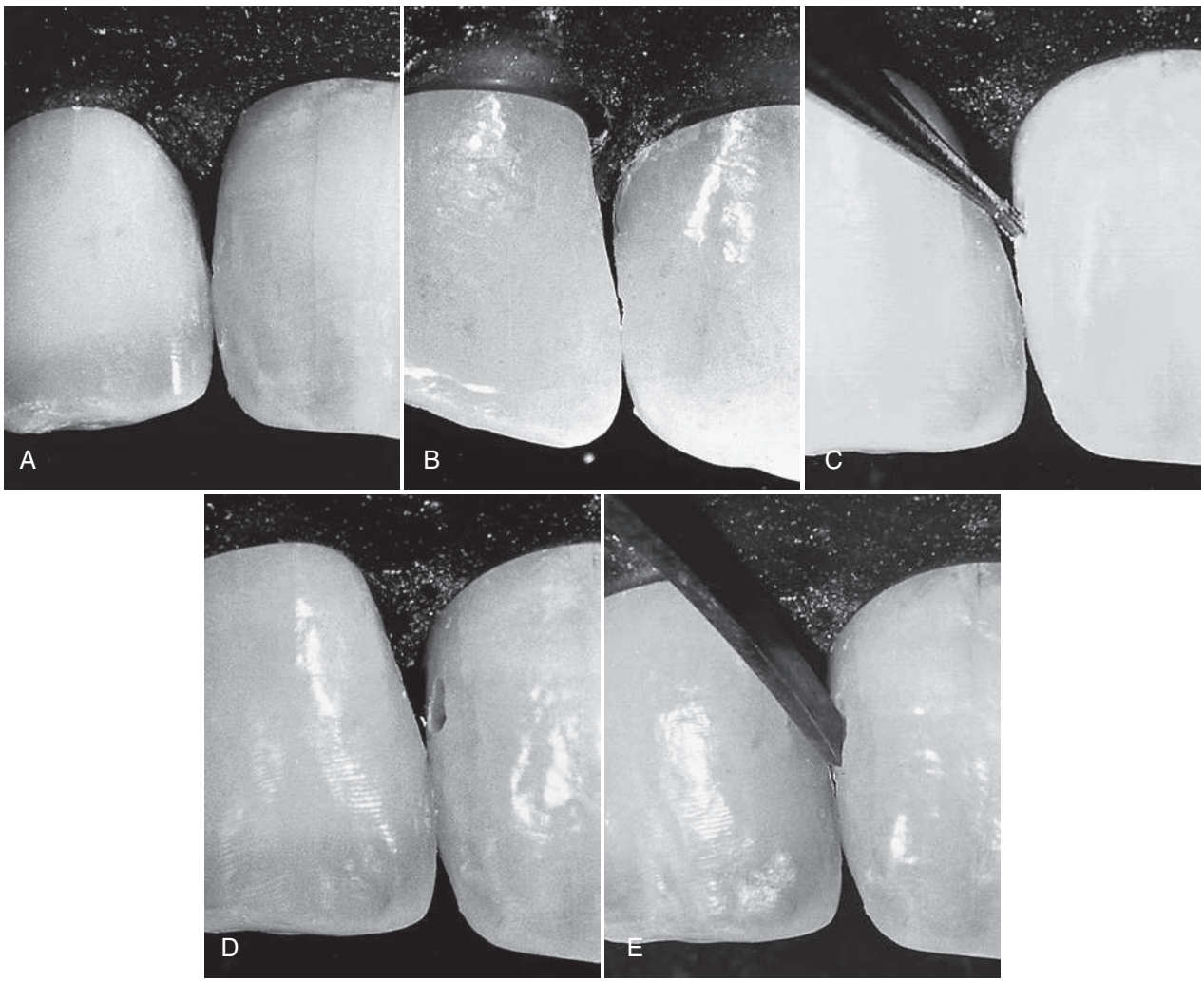
• **Fig. 17.36** Direct gold restoration of a clinical Class III preparation of straight-line design on the distal portion of the maxillary lateral incisor.



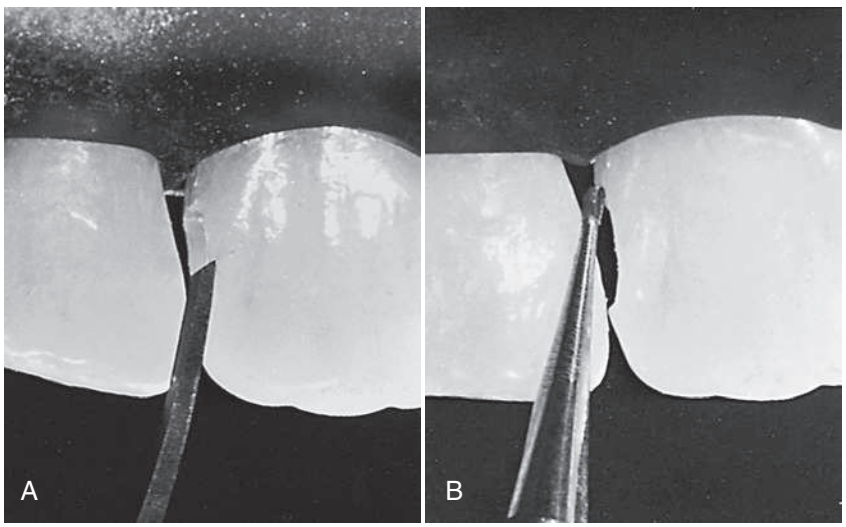
• **Fig. 17.38** Separator placed before clinical Class III preparation for the mandibular incisor.



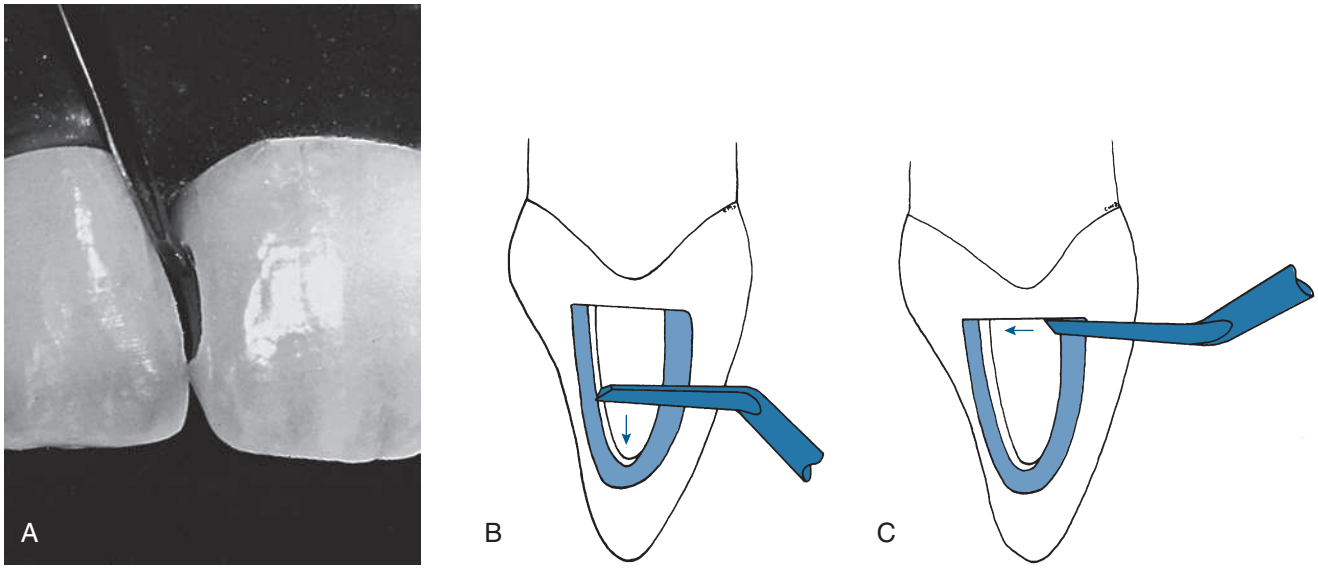
• **Fig. 17.37** Mandibular Class III preparation. A, Facial view. The facial margin is similar to that in the maxillary preparation. B, Linguoproximal view.



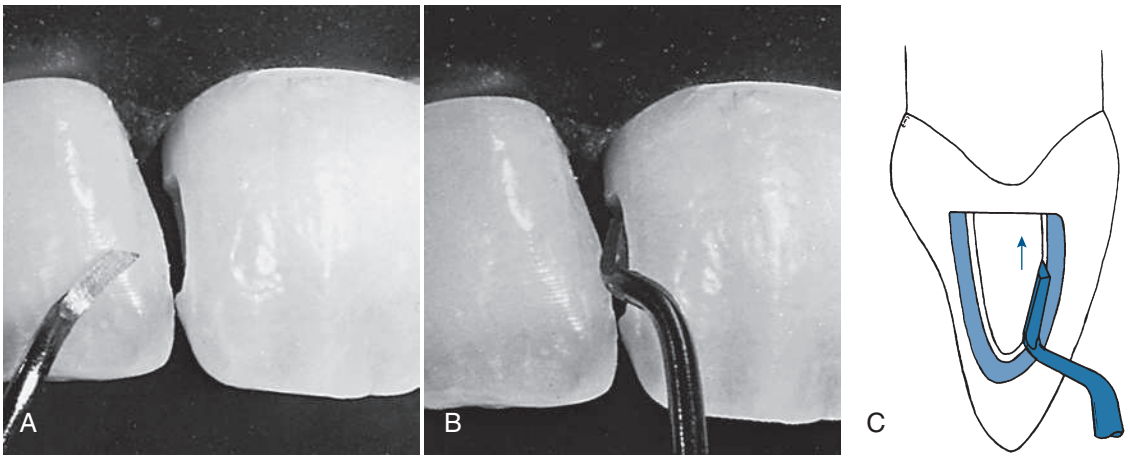
• **Fig. 17.39** A, Preoperative view of the extracted maxillary central incisor that has been mounted in dentoform. Distal surface to be treated with Class III cavity preparation and restoration of compacted gold. B, Preoperative lingual view. C, Facial approach initial entry is made with No. 33½ bur. D, Initial bur entry. E, The Wedelstaedt chisel begins to establish the facial outline form.



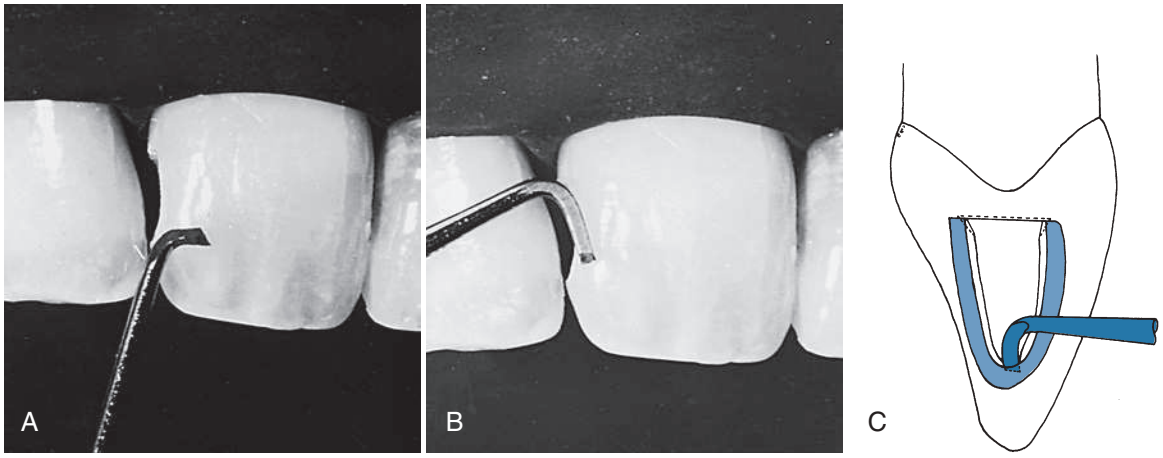
• **Fig. 17.40** Lingual view of preparation instrumentation. A, The Wedelstaedt chisel planing the lingual enamel wall. B, An inverted cone bur is used to establish the sharp linguogingival shoulder.



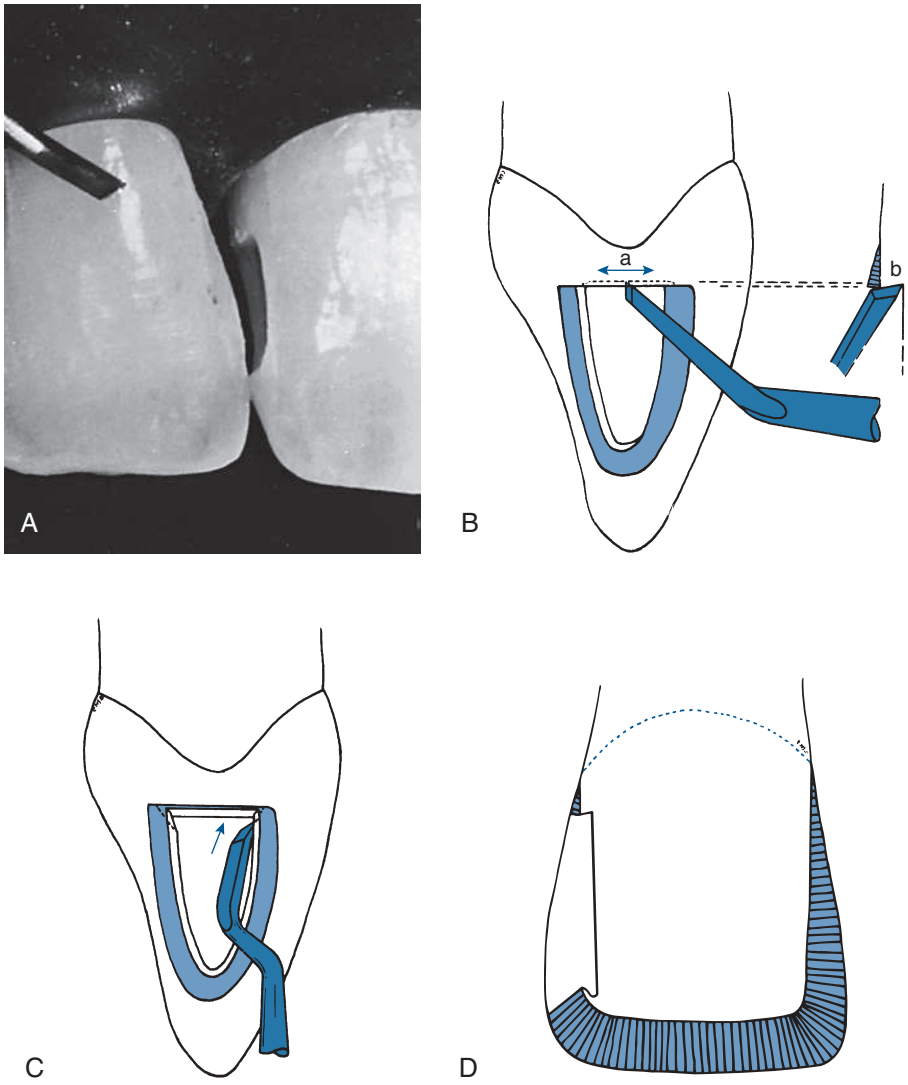
• **Fig. 17.41** Use of small hoe facial approach in tooth preparation. A, The hoe planes the lingual dentinal wall from the incisal aspect to the gingival aspect. B, The hoe also planes this wall from the gingival aspect to the incisal aspect *(arrow).* C, The hoe planes the gingival cavosurface *(arrow).*



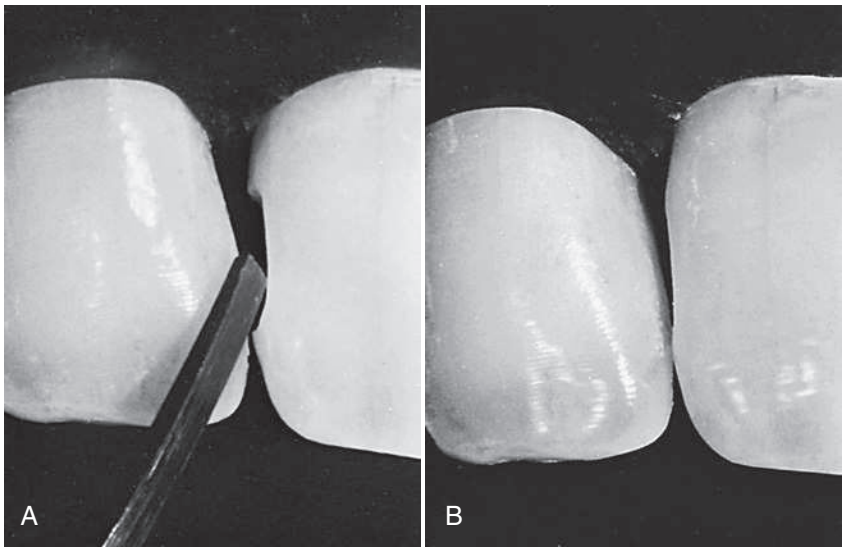
• **Fig. 17.42** Use of the angle former to plane the facial dentinal wall. A, Angle former before placement in the preparation. B, Angle former in the preparation. C, The angle former is directed apically *(arrow)* to plane the facial dentinal wall.



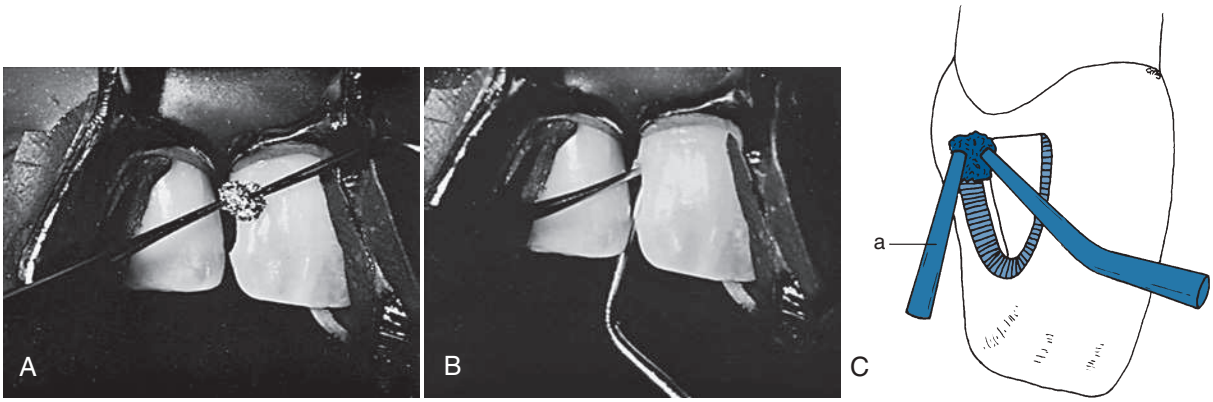
• **Fig. 17.43** A, Axial plane before placement in the preparation. B, Bibeveled hatchet before placement in the preparation. C, The bibeveled hatchet is used to establish the incisal retentive angle.



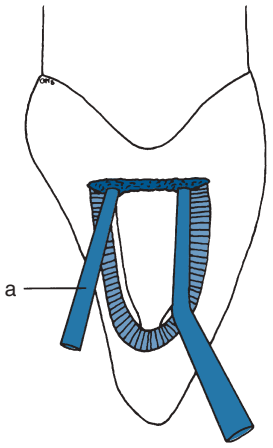
• **Fig. 17.44** A, Angle former before use in the preparation. B, The angle former is moved faciolingually *(a)* to establish an acute axiogingival line angle *(b)*. C, The offset angle former thrust faciogingivally establishes an acute facioaxiogingival point angle. D, Completed incisal, gingivoaxial retention form. (D, From Stibbs GD: Direct golds in dental restorative therapy. *Oper Dent* 5:107, 1980.)



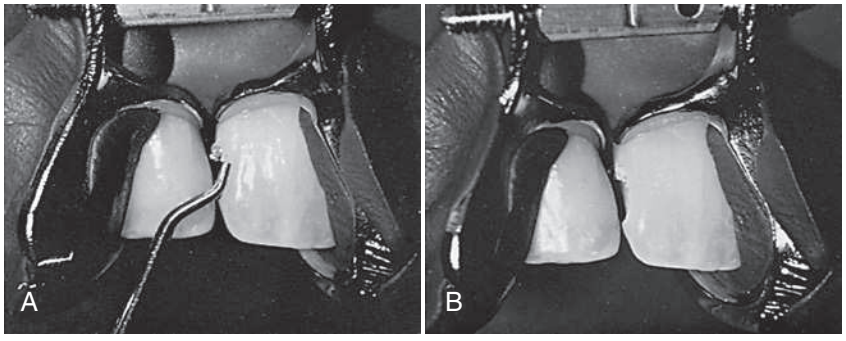
• **Fig. 17.45** A, The Wedelstaedt chisel may be used again to plane margins. B, Completed facial margin of Class III tooth preparation viewed from the facial position.



• **Fig. 17.46** A, The first pellet of the gold foil is placed from the facial aspect into the preparation. Note the separation of teeth by 0.25 to 0.5 mm. B, Compaction of the pellet into the linguoaxiogingival point angle. The line of force is directed linguoaxiogingivally, while the holding instrument is placed from the lingual position. C, The holding instrument *(a)* prevents dislodgment of foil during compaction.



• **Fig. 17.47** The holding instrument *(a)* remains in position as the gold foil is condensed across the gingival wall toward the facial portion of the preparation.

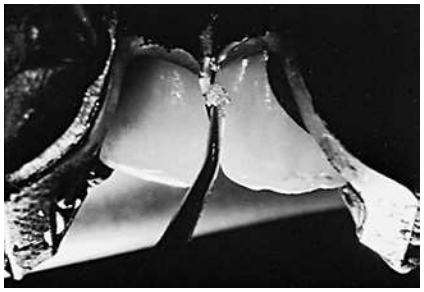


• **Fig. 17.48** A, Offset condenser before placement in the cavity preparation. B, Compacted gold foil covering the gingival wall and the cavosurface.

that the incisal surface of the growing restoration always slope apically, with the gold on the axial wall ahead of the proximal surface of the restoration. During the compaction procedure, the vector of the line of force always should be toward the internal portion of the preparation to prevent dislodgment of the restoration.

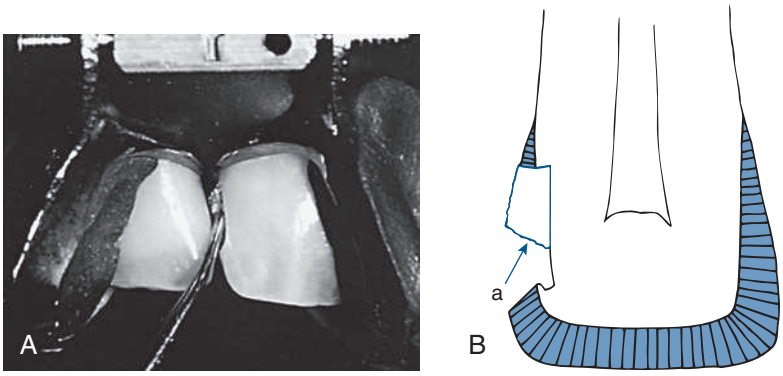
The next step is the restoration of the incisal portion of the preparation, referred to as "making the turn." It is accomplished in three phases. First, sufficient gold is built up on the lingual wall so that the gold is near the incisal angle (Fig. 17.51). Second, the incisal area is filled by compacting $\frac{1}{128}$-size pellets with the right-angle hand condenser (Fig. 17.52). Third, pellets of foil are compacted into the incisolingual and incisal areas with the offset condenser. This fills the incisal portion, making a complete turn from lingual to facial (Fig. 17.53A). The entire incisal cavosurface is covered with gold (see Fig. 17.53B).

Additional gold compaction finishes the facial one third of the restoration, and then the Varney foot condenser is used to

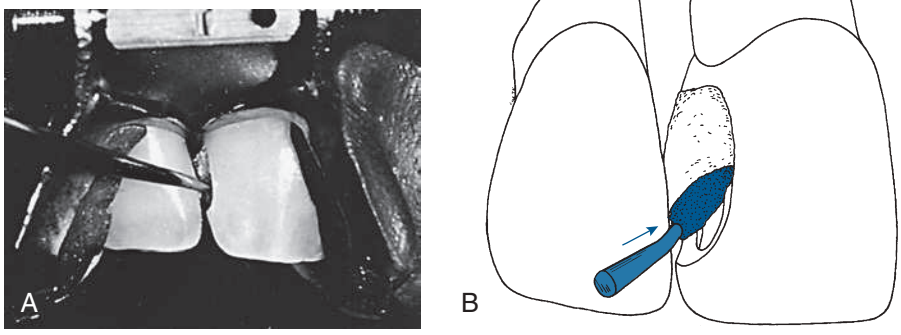


• **Fig. 17.49** Lingual view. The monangle condenser confirms compaction of gold at the linguogingival aspect of the restoration.

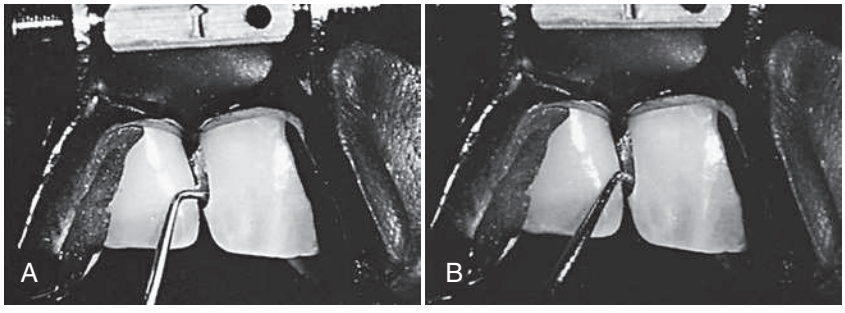
"after-condense" over the contour of the restoration. More separation is generated by slight activation of the separator, before finishing and polishing the restoration. A sharp, gold foil knife is used to remove excess in the contact area, permitting a fine finishing strip or steel matrix strip to pass through. A pull-cut Shooshan file or gold knife may facilitate removal of excess gold facially (Fig. 17.54). Initial contouring of the contact area is performed with long, extra-narrow, extra-fine cuttle finishing strips to gain access to the proximal surface. Next, a wide, medium cuttle strip may be used for rapid removal of excess gold. Final contouring continues with the medium and fine, narrow strips. Finishing is performed with the extra-narrow, extra-fine cuttle strip



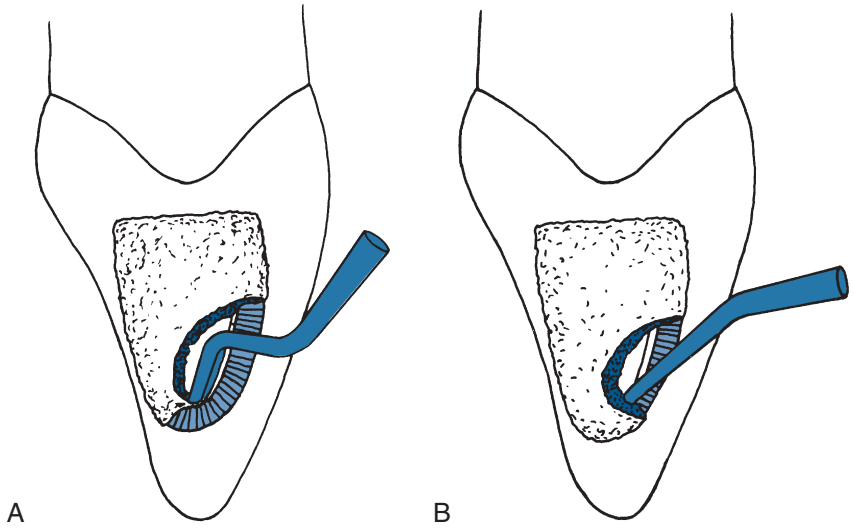
• **Fig. 17.50** A, The monangle condenser is used to build the bulk of gold in the gingival half of the preparation. B, Gingival half of the restoration in longitudinal section. The line of force *(a)* is directed axiogingivally during compaction of gold to prevent dislodgment of the restoration.



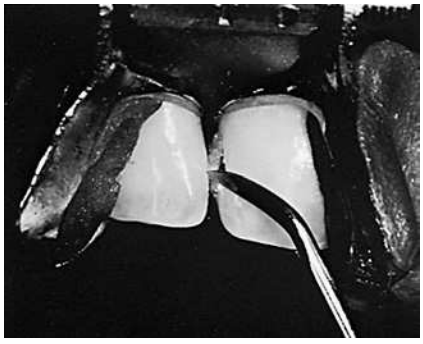
• **Fig. 17.51** A, The condenser is directed over the facial surface of the adjacent tooth, while the gold is built toward the incisal aspect. B, The gold is compacted from the facioincisal aspect to cover the lingual cavosurface; however, compaction direction must continue to have a major vector *(arrow)* toward the axial wall to prevent dislodgment. At this stage, the compacted foil on the axial wall must be well ahead (incisally) of the "growing" proximal surface.



• **Fig. 17.52** A, The right-angle hand condenser begins to press the gold into the incisal retention. B, This condenser forces the gold deeply into the incisal retentive undercut.



• **Fig. 17.53** Completing the compaction of gold into the incisal region of the preparation. A, The offset bayonet condenser condenses the gold into the incisal retention with mallet compaction. B, The incisal cavosurface is restored with gold foil condensed with the small monangle condenser.



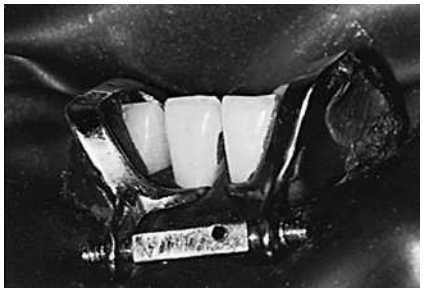
• **Fig. 17.54** A sharp, thin-bladed gold knife removes excess gold from the facial surface.



• **Fig. 17.55** Fine cuttle finishing strips polish the proximal surface of the gold foil restoration.



• **Fig. 17.56** Completed maxillary Class III gold foil restoration.



• **Fig. 17.57** Completed mandibular Class III gold foil restoration.

(Fig. 17.55). Care is taken to finish only the facial or lingual areas with the strip and to avoid flattening the contact area. The gold knife or discoid-cleoid instrument can be used to remove the final excess gold from the cavosurface margins. The separator is then removed. Final polishing is accomplished with a worn, extra-fine cuttle strip. Polishing powder may be used. Omitting this step results in a satin finish that is less reflective of light and perhaps more esthetically pleasing (Fig. 17.56).

Summary

Direct-filling gold is useful in restorative dentistry. If carefully manipulated by a dentist, this restorative material may provide lifetime service to patients and promote their oral health (Fig. 17.57). Direct-filling gold contributes to the art and the science of restorative dentistry.

References

1. Dwinelle WH: Crystalline gold, its varieties, properties, and use. *Am J Dent* 5:249, 1855.
2. Ferrier WI: The use of gold foil in general practice. *J Am Dent Assoc* 28:691, 1941.
3. Hollenback GM: There is no substitute for gold foil in restorative dentistry. *J South Calif Dent Assoc* 33:275, 1965.
4. Lambert RL: A survey of the teaching of compacted gold. *Oper Dent* 5:20, 1980.
5. Stibbs GD: Direct golds in dental restorative therapy. *Oper Dent* 5:107, 1980.
6. Trueman WH: An essay upon the relative advantage of crystallized gold and gold foil as a material for filling teeth. *Dent Cosmos* 10:128, 1868.
7. Ingersol CE: Personal communication, 1982.
8. Lund MR, Baum L: Powdered gold as a restorative material. *J Prosthet Dent* 13:1151, 1963.
9. Hodson JT: Structure and properties of gold foil and mat gold. *J Dent Res* 42:575, 1963.
10. Hodson JT: Compaction properties of various pure gold restorative materials. *J Am Acad Gold Foil Oper* 12:52, 1969.
11. Smith GE: *The effect of condenser design and lines of force on the dental compaction of cohesive gold [Master's thesis]*, Seattle, 1970, University of Washington.
12. Black GV: The nature of blows and the relation of size of plugger points force as used in filling teeth. *Dent Rev* 21:499, 1907.
13. Baum L: Gold foil (filling golds) in dental practice. *Dent Clin North Am* 199:1965.
14. Ivoclar-Williams Company: E-Z Gold instructional brochure, Amherst, NY, Ivoclar-Williams.
15. Smith GE: Condenser selection for pure gold compaction. *J Am Acad Gold Foil Oper* 15:53, 1972.
16. Hodson JT, Stibbs GD: Structural density of compacted gold foil and mat gold. *J Dent Res* 41:339, 1962.
17. Thomas JJ, Stanley HR, Gilmore HW: Effects of gold foil condensation on human dental pulp. *J Am Dent Assoc* 78:788, 1969.
18. Ferrier WI: Treatment of proximal cavities in anterior teeth with gold foil. *J Am Dent Assoc* 21:571, 1934.
19. Smith GE, Hodson JT, Stibbs GD: A study of the degree of adaptation possible in retention holes, convenience points and point angles in Class III cavity preparations. *J Am Acad Gold Foil Oper* 15:12–18, 1972.